

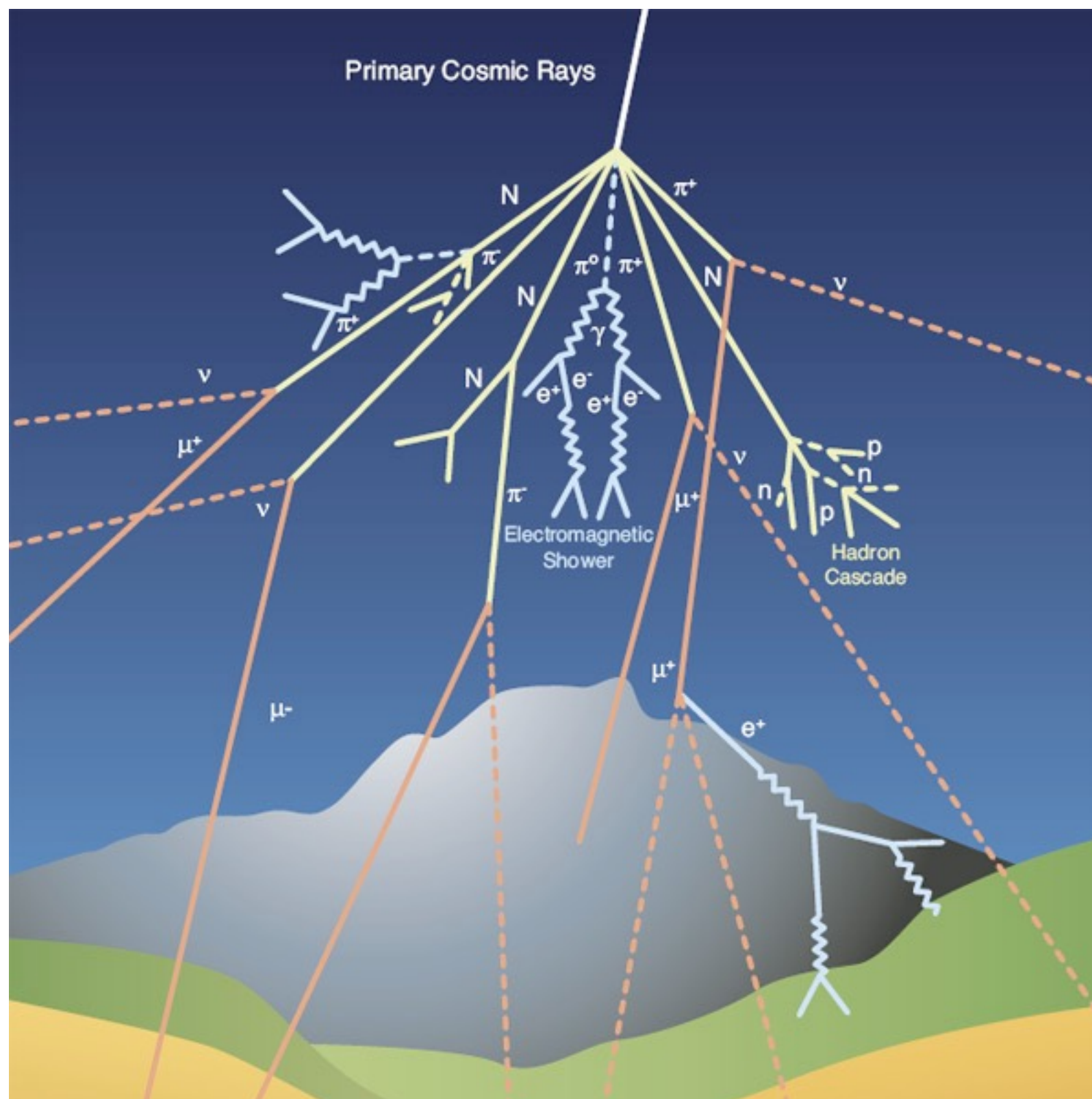


Astrophysical cosmic rays lecture IV

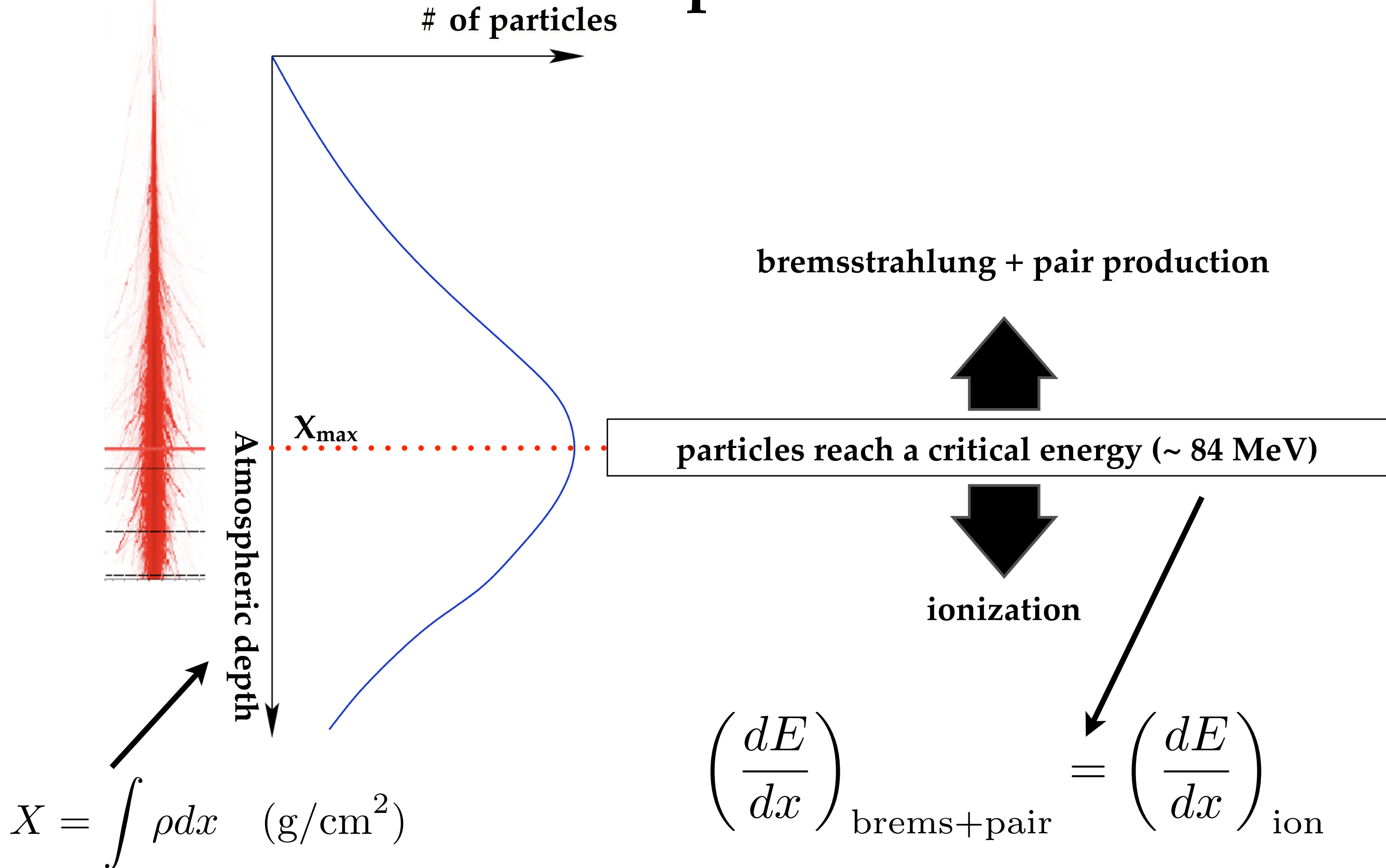
Extensive air showers and experimental techniques

Edivaldo Moura Santos
Instituto de Física - USP

Extensive Air Shower (EAS)

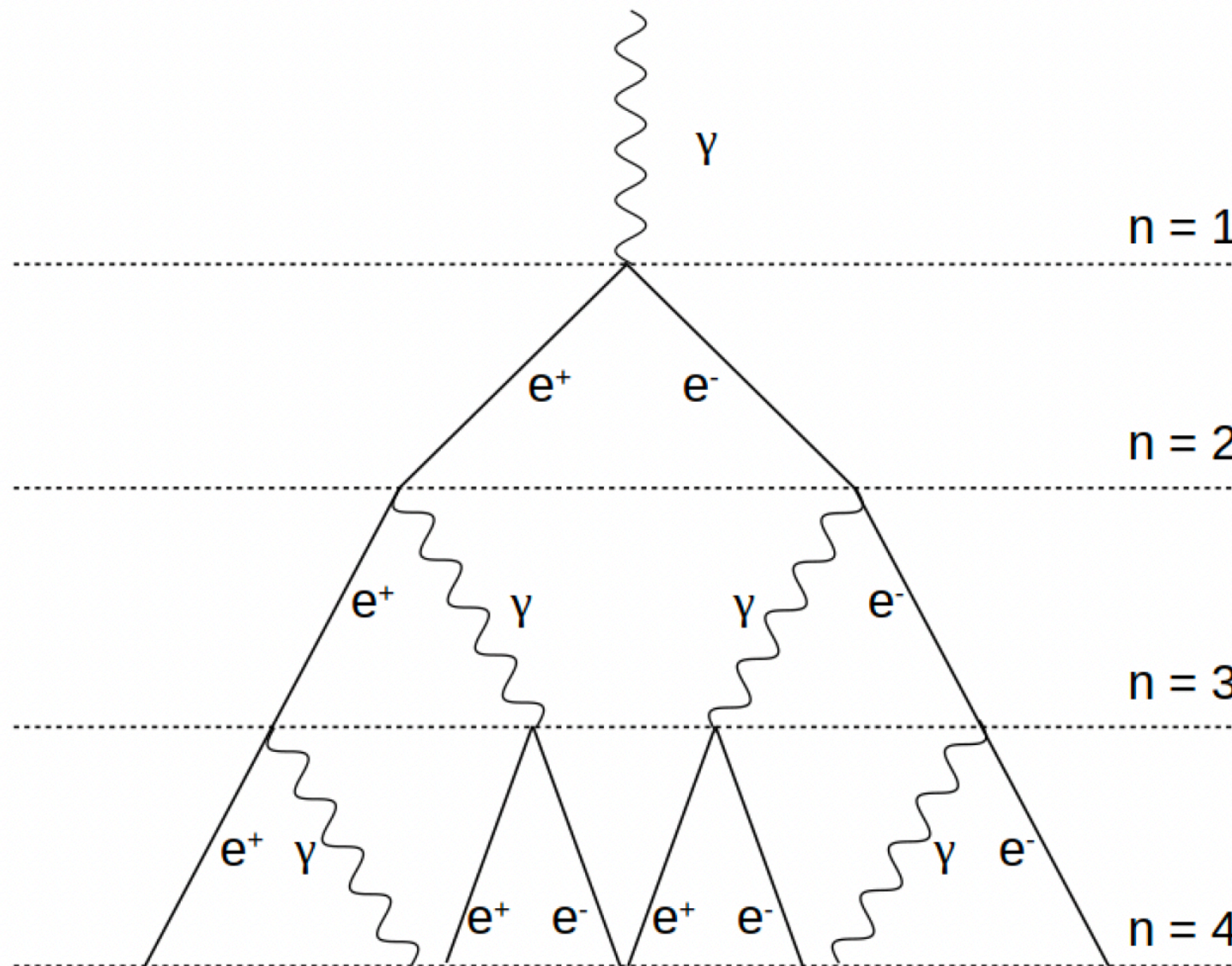


Electromagnetic shower longitudinal development



The Heitler model

- Heitler developed a simple model for the evolution of an electromagnetic shower, that is, one formed exclusively by electrons, positrons and photons.
- His idea was to approximate this shower using a sequence of binary splittings as represented below



Walter Heitler. The Quantum Theory of Radiation. Oxford At The Clarendon Press, 1 edition, 1936.

The Heitler model

- In Heitler's model, the average atmospheric depth traveled by a photon or electron/positron before interacting is

$$d = \lambda_r \ln 2$$

$$\lambda_r \simeq 37 \text{ g cm}^{-2}$$

radiation length

- If the first interaction happens at a depth X_0 , then at a depth X , the number of binary splittings n (i.e., particles in the shower) is given by

$$n = \frac{X - X_0}{d} \implies N(X) = 2^{(X - X_0)/d}$$

- The average energy per particle at depth X

$$E(X) = \frac{E_0}{N(X)}$$

- The number of particles at the depth where the shower reaches its maximum development can be written in terms of the primary particle energy E_0 and the critical energy E_c

$$N(X_{max}) = \frac{E_0}{E_c} \qquad E_c \simeq 84 \text{ MeV}$$

The Heitler model

- The 3 main predictions of Heitler's model for electromagnetic cascades can be summarized as

1. The number of particles at the depth of maximum development $X_{\max}$ is proportional to the primary particle's energy

$$N(X_{max}) = \frac{E_0}{E_c}$$

2. The energy evolution of $X_{\max}$ is logarithmic

$$X_{max} = X_0 + \lambda_r \ln \left(\frac{E_0}{E_c} \right)$$

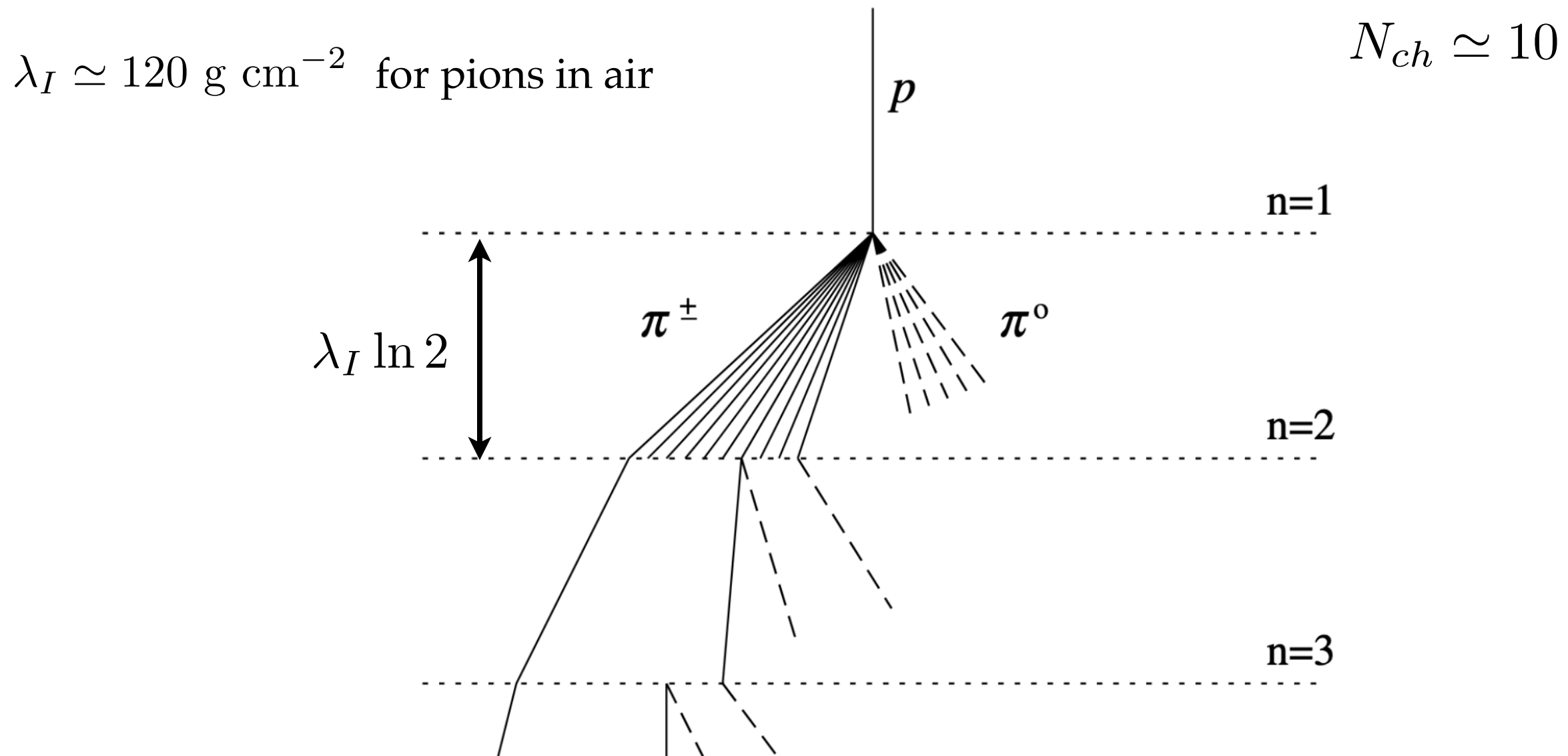
3. The so called elongation rate is completely defined by the radiation length

$$D_{10} \equiv \frac{dX_{max}}{d \log_{10} E_0} = 2.3 \lambda_r$$

- As a first approximation, these predictions are verified by more detailed shower simulations. On the other hand, we know that more photons can be emitted via bremsstrahlung, so electrons lose energy faster than photons. As a consequence, the Heitler model tend to overestimate the number of particles at $X_{\max}$ by a factor 2 to 3.

The Heitler-Matthews model

- J. Matthews extended Heitler's electromagnetic shower model to describe hadronic showers
- In his model, at each interaction layer along the atmosphere, a total of N_{ch} charged pions are created together with $N_{ch}/2$ neutral pions, the latter decaying immediately into 2 photons and feeding the electromagnetic cascade.



J. Matthews. A Heitler model of extensive air showers. Astroparticle Physics, 22:387–397, jan 2005.

The Heitler-Matthews model

- The interaction length λ_I is assumed to be constant (a good approximation for interactions in the energy range 10-1000 GeV)
- Charged pion interactions proceed until they fall below the critical energy where the decay probability is larger than the interaction one. In fact, in Heitler-Matthews model, all the charged pions are assumed to decay below the critical energy

$$E_c^\pi \simeq 20 \text{ GeV}$$

- In fact, the critical energy should be slightly dependent on the energy of the primary particle, but in Heitler-Matthews model, it is taken as a fixed value.
- Charged pion decays create a muon sub-cascade

$$\pi \longrightarrow \mu + \nu$$

The Heitler-Matthews model

- After n interactions (i.e. interaction layers), the total number of charged pions is given by

$$N_{\pi} = (N_{ch})^n$$

- Assuming the energy is uniformly shared among the pions, the average energy per charged pion after n interactions is

$$E_{\pi} = \frac{\left(\frac{2}{3}\right)^n E_0}{(N_{ch})^n}$$

- **TASK:** show that the number of interaction layers n_c necessary for a charged pion in the HM model to reach the critical energy E_c^{π} is given by

$$n_c = \frac{\ln(E_0/E_c^{\pi})}{\ln(\frac{3}{2}N_{ch})} \simeq 0.85 \log_{10} \left(\frac{E_0}{E_c^{\pi}} \right)$$

- Summary of HM model input parameters:

$$\left\{ \begin{array}{l} N_{ch} = 10 \\ E_c^{em} = 84 \text{ MeV} \\ E_c^{\pi} = 20 \text{ GeV} \\ \lambda_r = 37 \text{ g cm}^{-2} \\ \lambda_I = 120 \text{ g cm}^{-2} \end{array} \right.$$

The Heitler-Matthews model

- At the depth of maximum development, the energy of the primary particle is divided between $N_\pi (= N_\mu)$ pions and $N_{\max}$ electromagnetic particles in the sub showers created by neutral pion decays. So, we can write:

$$E_0 = E_c^{em} N_{max} + E_c^\pi N_\mu$$

- Writing the number of electromagnetic particles in terms of the number of electrons

$$N_e = \frac{N_{max}}{g} \quad \text{with} \quad g \simeq 10$$

- We can finally write

$$E_0 = g E_c^{em} \left(N_e + \frac{E_c^\pi}{g E_c^{em}} N_\mu \right) \simeq 0.85 (N_e + 24 N_\mu) [\text{GeV}]$$

- See that it should be possible to estimate the energy of a cascade by sampling its electromagnetic and muon component at ground level.

The Heitler-Matthews model

- At the depth of maximum development, the energy of the primary particle is divided between $N_\pi (= N_\mu)$ pions and N_{max} electromagnetic particles in the sub showers created by neutral pion decays. So, we can write:

$$E_0 = E_c^{em} N_{max} + E_c^\pi N_\mu$$

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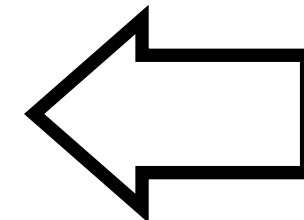
$$E_0 = g E_c^{em} \left(N_e + \frac{E_c^\pi}{g E_c^{em}} N_\mu \right) \simeq 0.85 (N_e + 24 N_\mu) [\text{GeV}]$$

- See that it is possible to estimate the energy of a cascade by sampling its electromagnetic and muon component at shower maximum. In practice, different detectors do this at different stages of shower development.

Shower simulations

- Even though, Heitler and Heitler-Matthews models are useful to get an intuition on how an atmospheric cascade develops, both these models are based on a series of simplified assumptions.
- With the advance on the processing and storage capabilities of computers, it is nowadays possible to simulate showers, using appropriate algorithms of sampling, up to energies of 10^{21} eV.
- Therefore, current analyses of EAS properties are done with softwares like CORSIKA (COsmic Ray SIMulations for KAscade) :

<https://www.iap.kit.edu/corsika/>

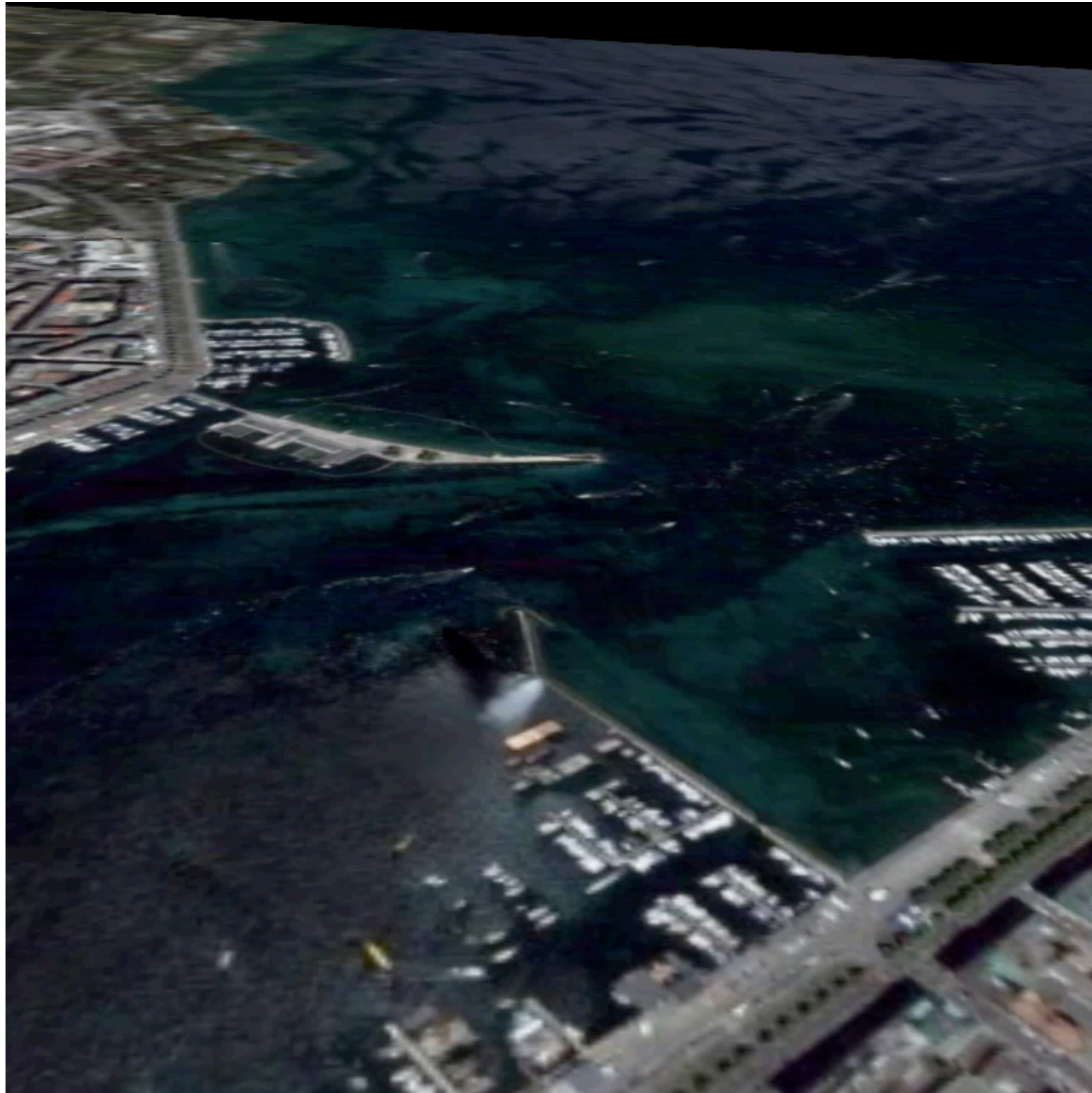


State of the art

- or AIRES (AIR shower Extended Simulations)

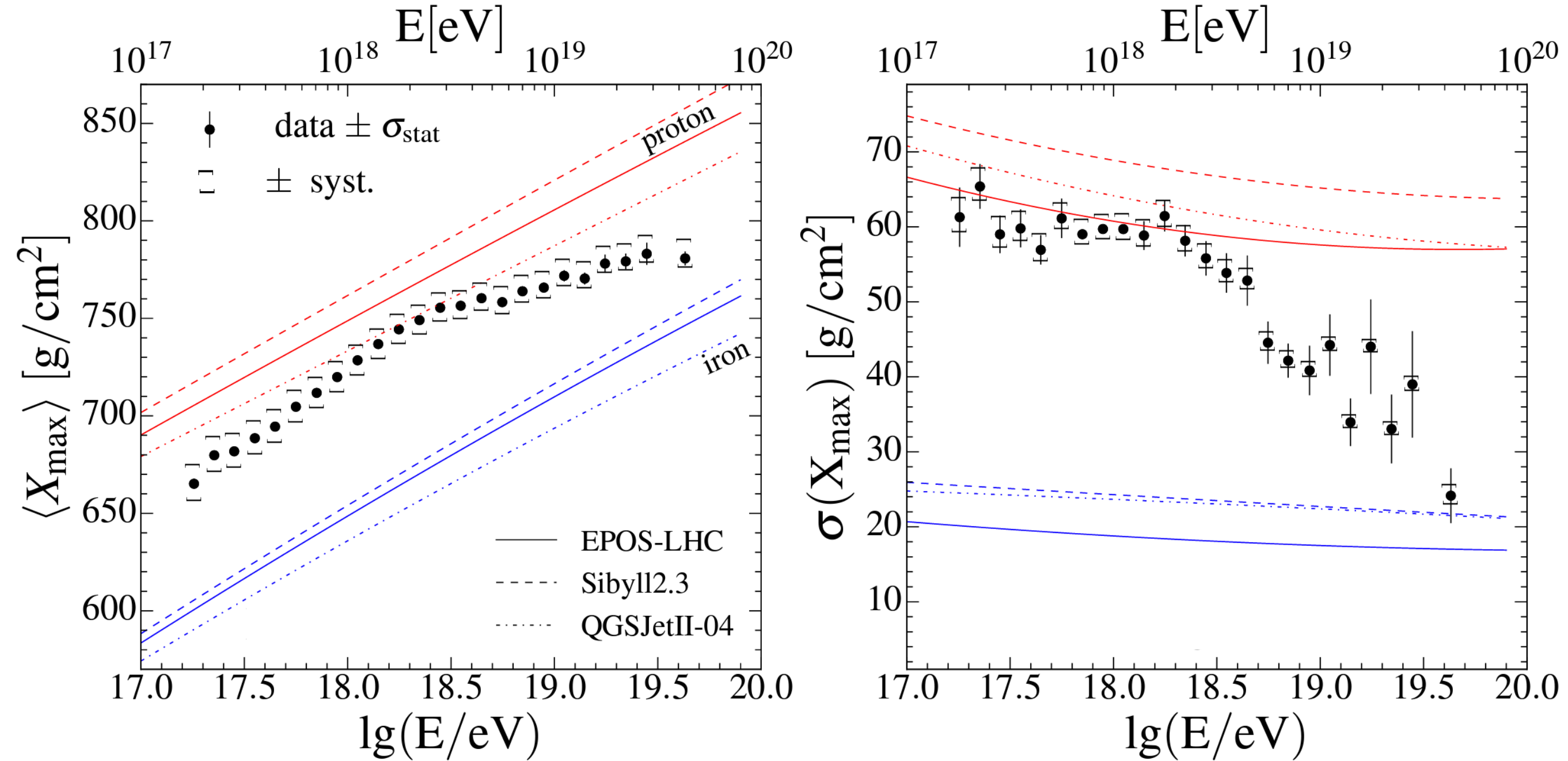
<http://aires.fisica.unlp.edu.ar/>

EAS simulation



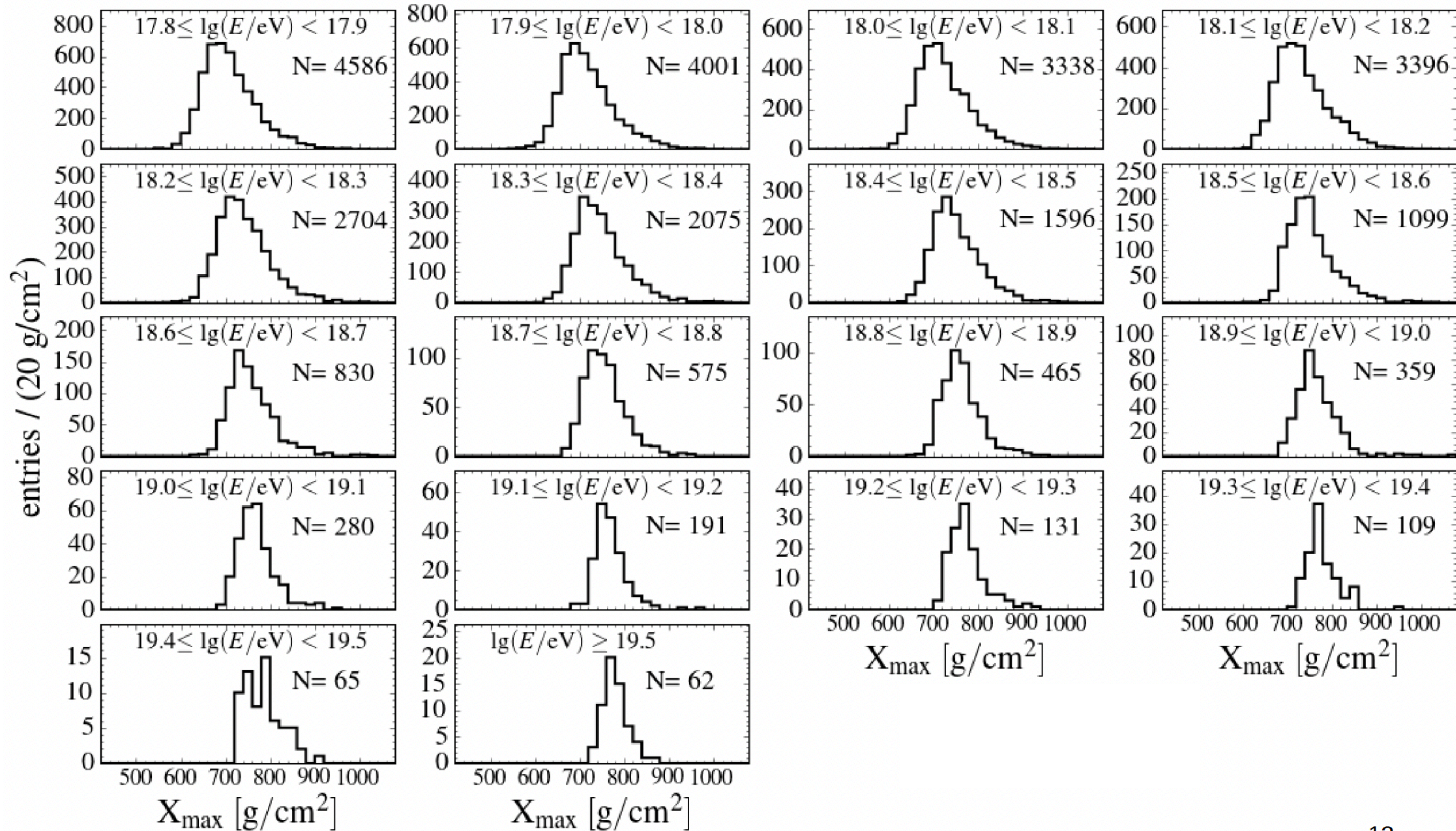
COSMUS group : <http://astro.uchicago.edu/>

$X_{\max}$ and its fluctuations



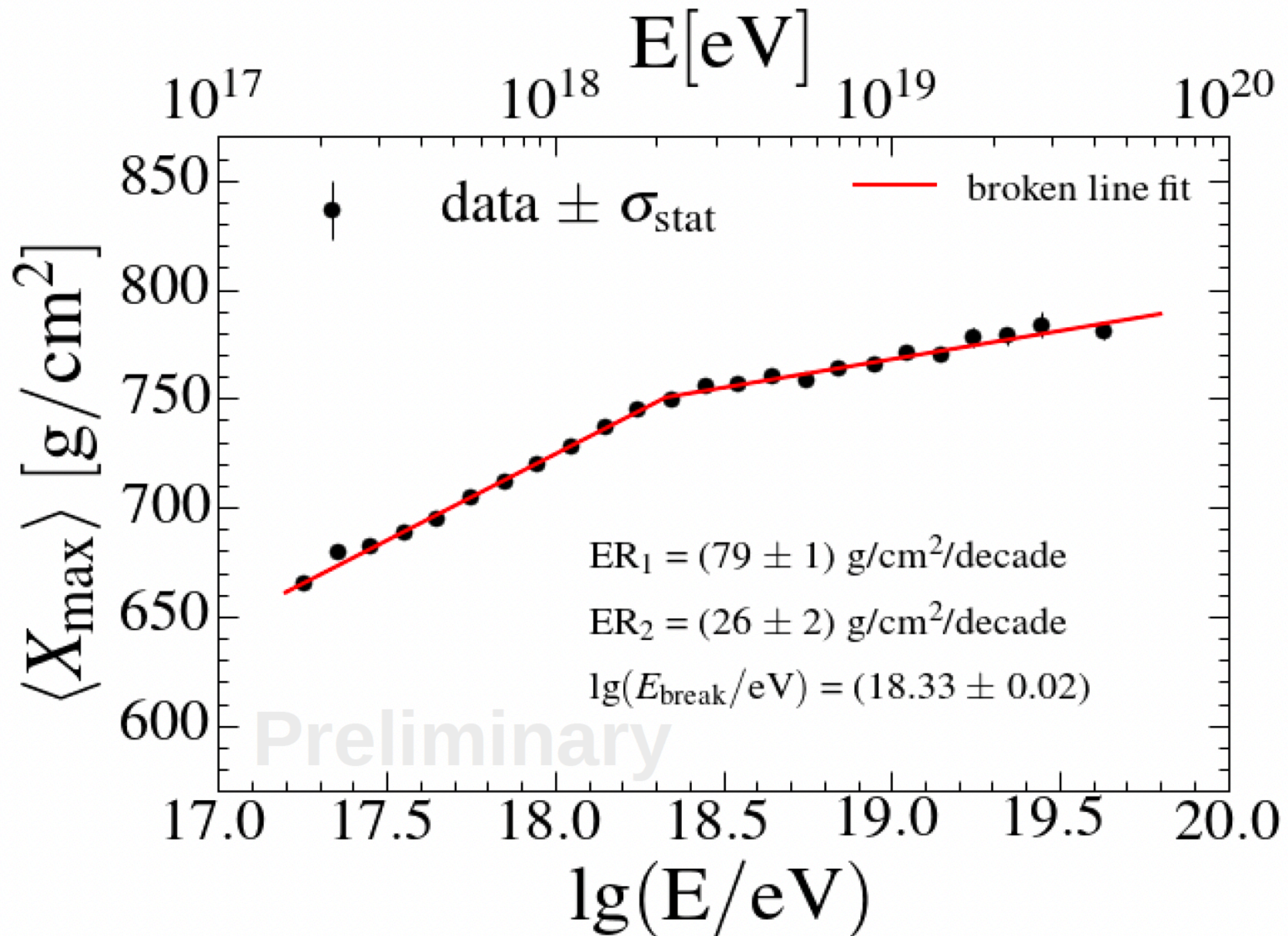
The Pierre Auger Collaboration, Depth of maximum of air-shower profiles at the Pierre Auger Observatory: Measurements above $10^{17.2}$ eV and Composition Implications, 35th ICRC 2017.

$X_{\max}$ and its fluctuations



The Pierre Auger Collaboration, Depth of maximum of air-shower profiles at the Pierre Auger Observatory: Measurements above $10^{17.2} \text{ eV}$ and Composition Implications, 35th ICRC 2017.

Elongation rate



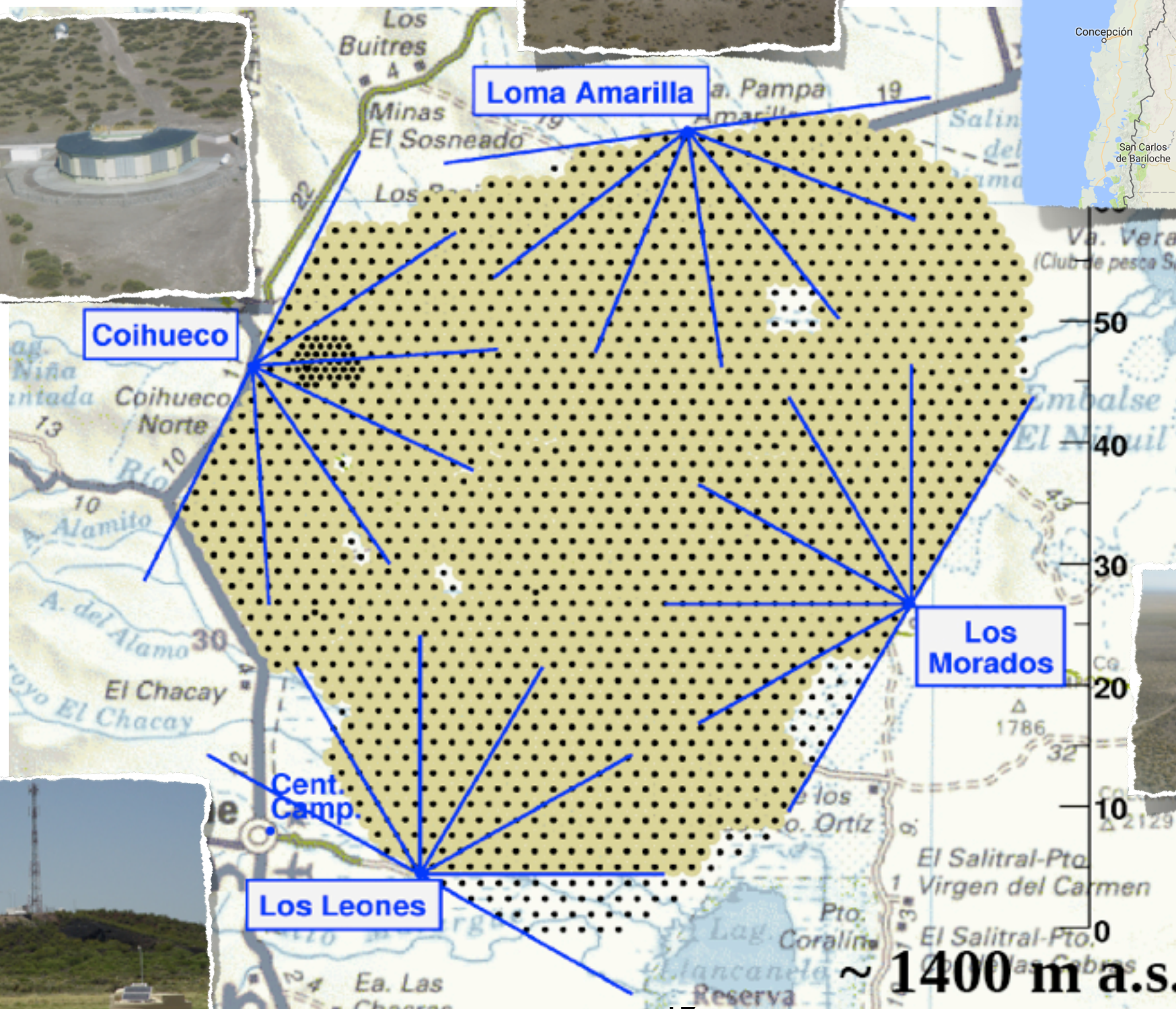
The Pierre Auger Collaboration, Depth of maximum of air-shower profiles at the Pierre Auger Observatory: Measurements above $10^{17.2}$ eV and Composition Implications, 35th ICRC 2017.

The Pierre Auger Observatory

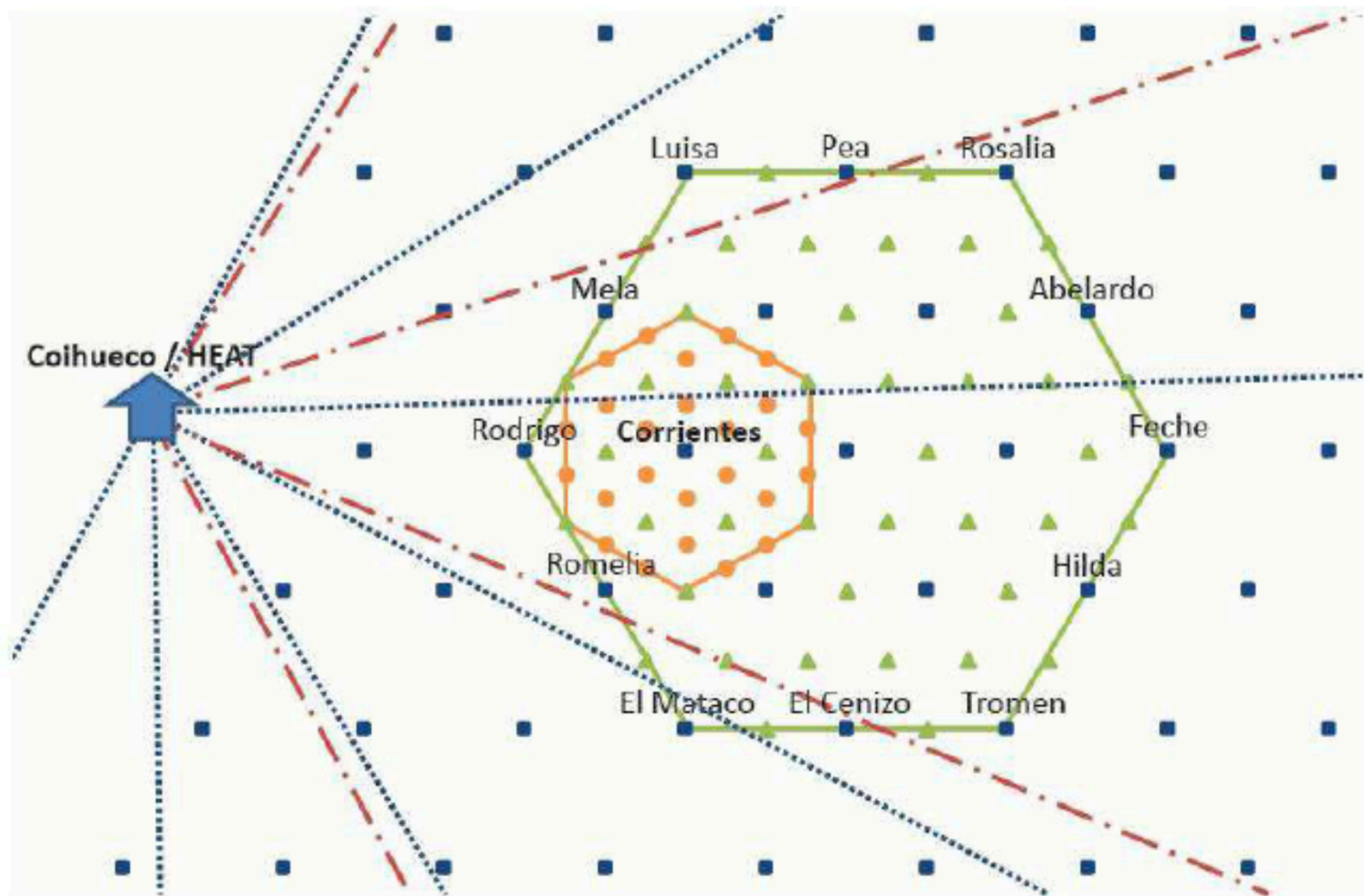
3000 km² surface array (1660 detectors)

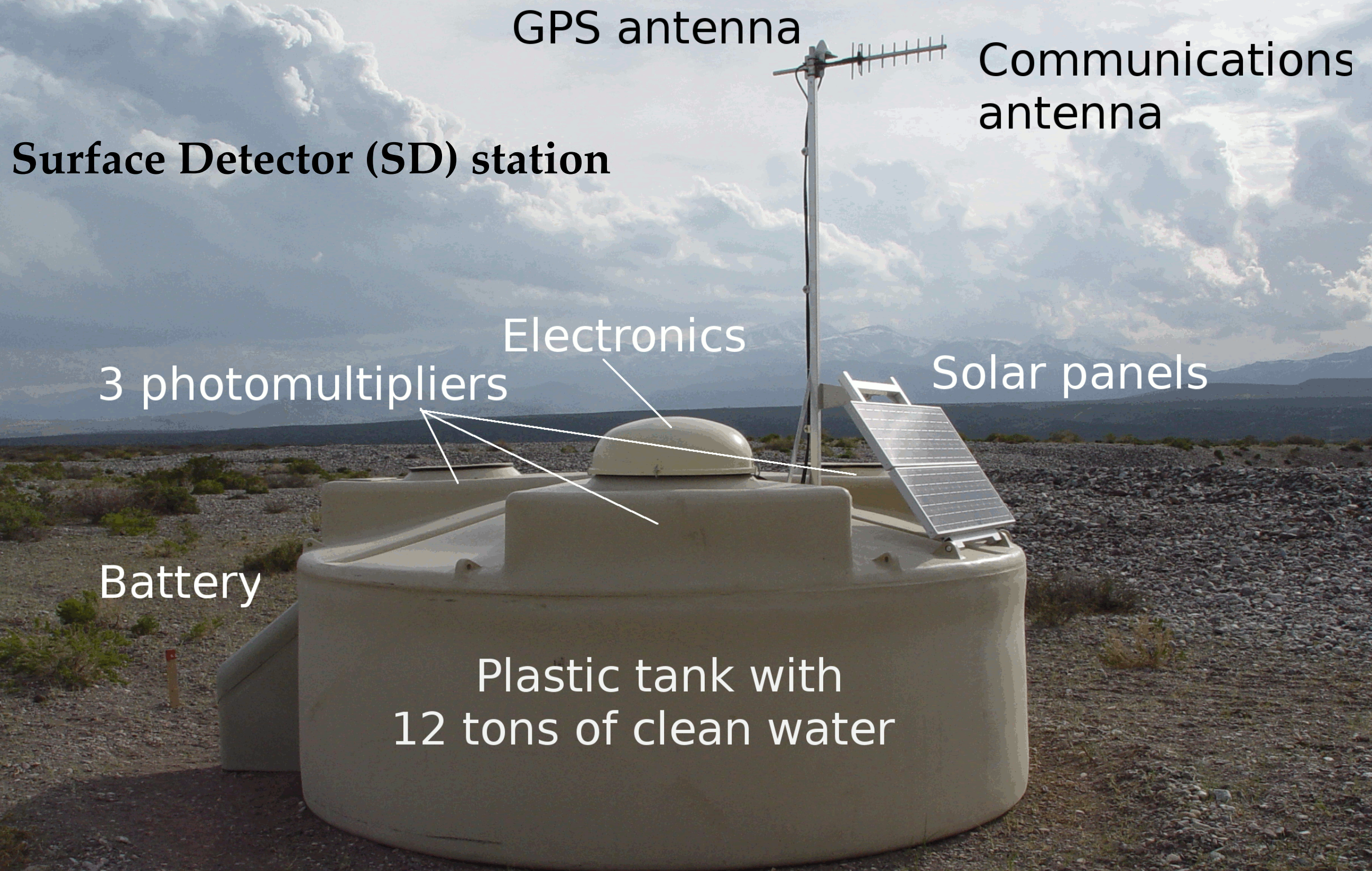
1.5 km triangular grid

27 fluorescence telescopes



The AMIGA infill

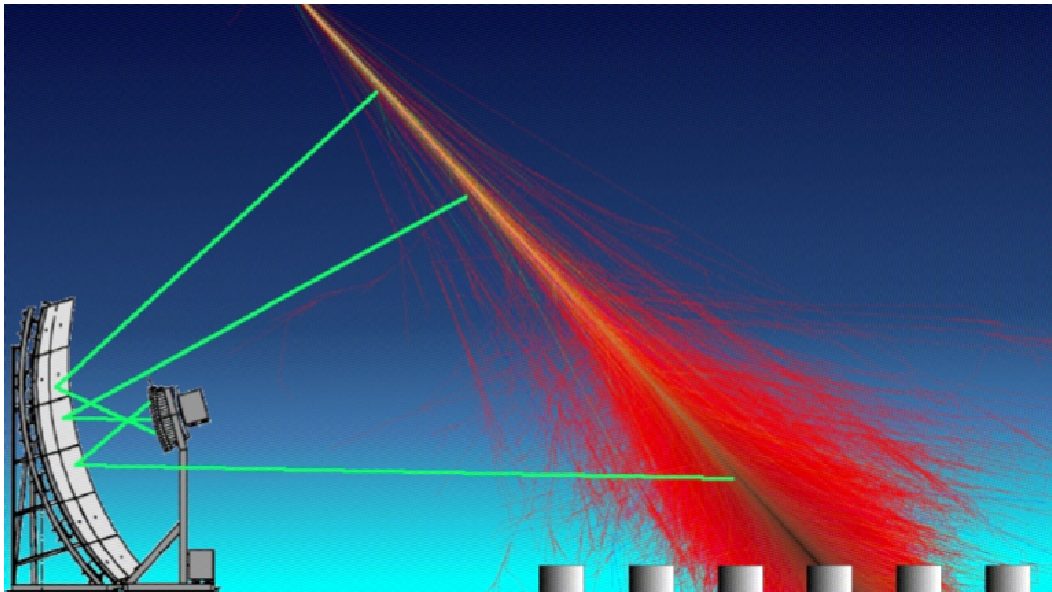
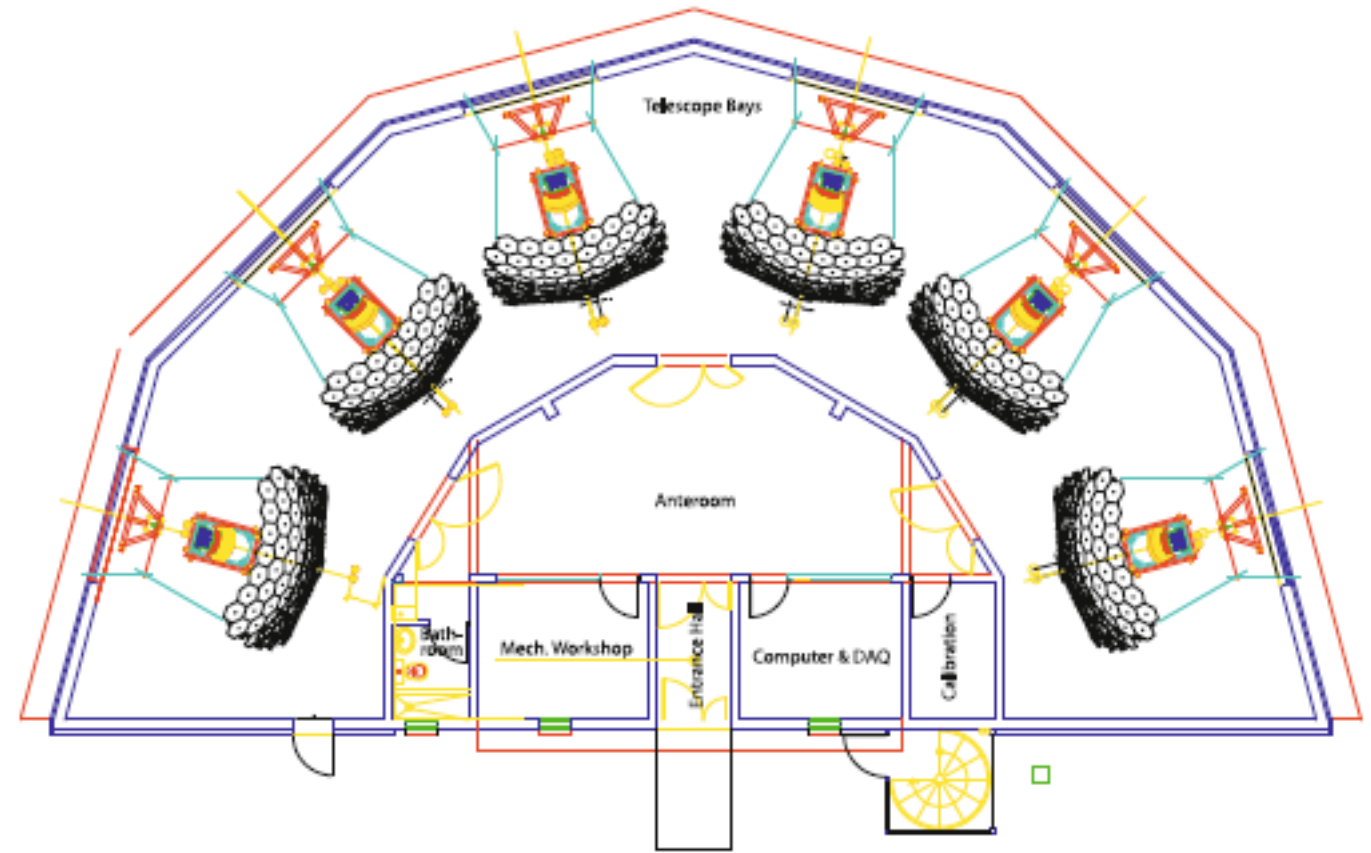




Loma Amarilla

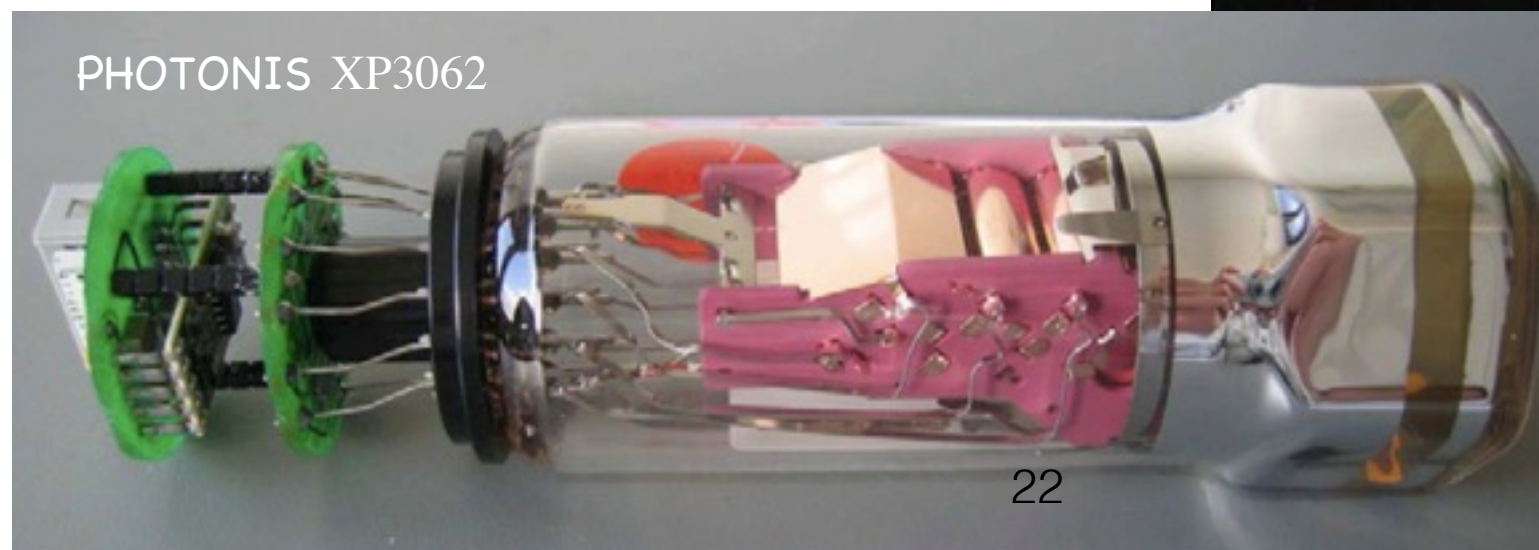
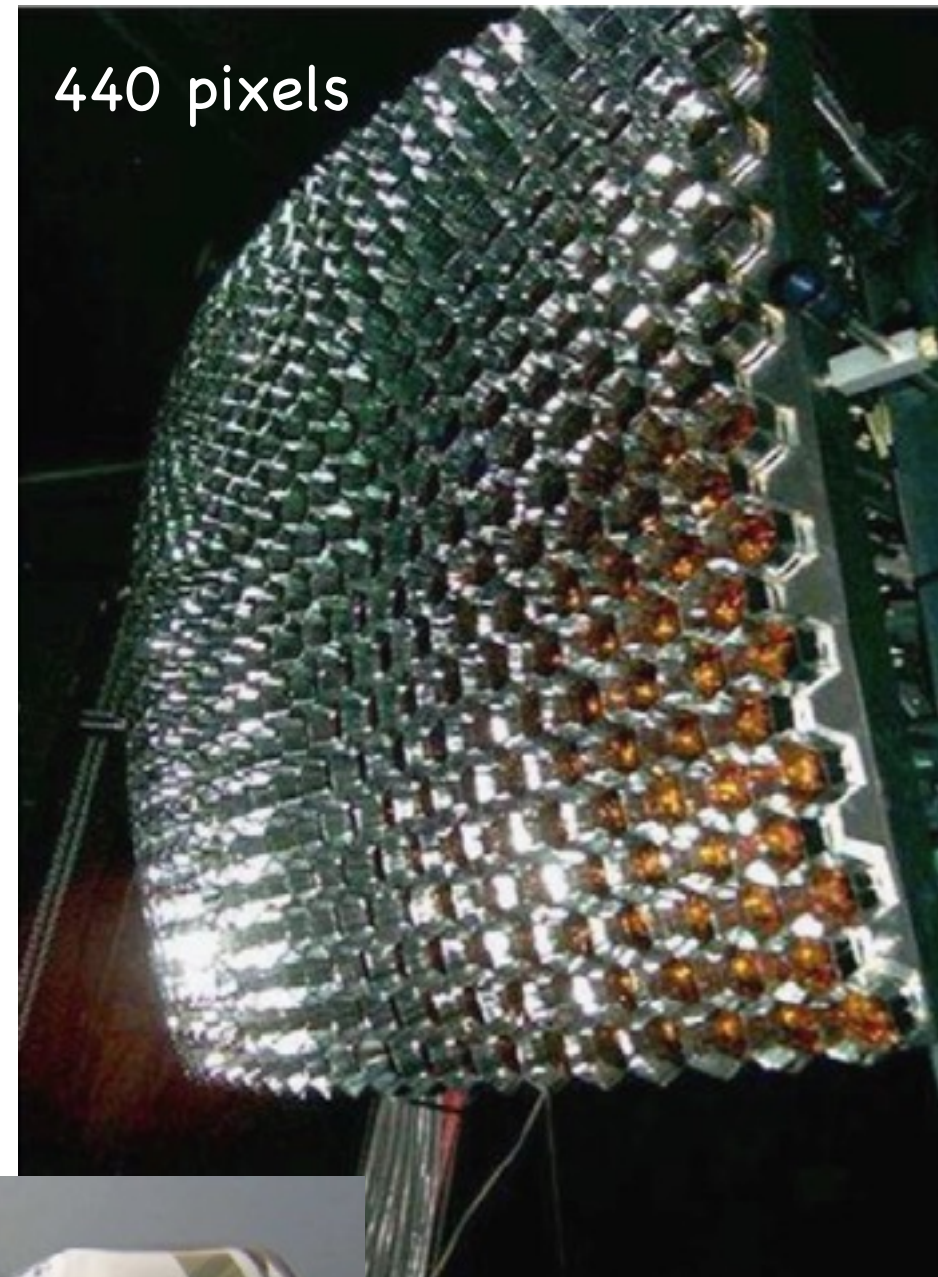
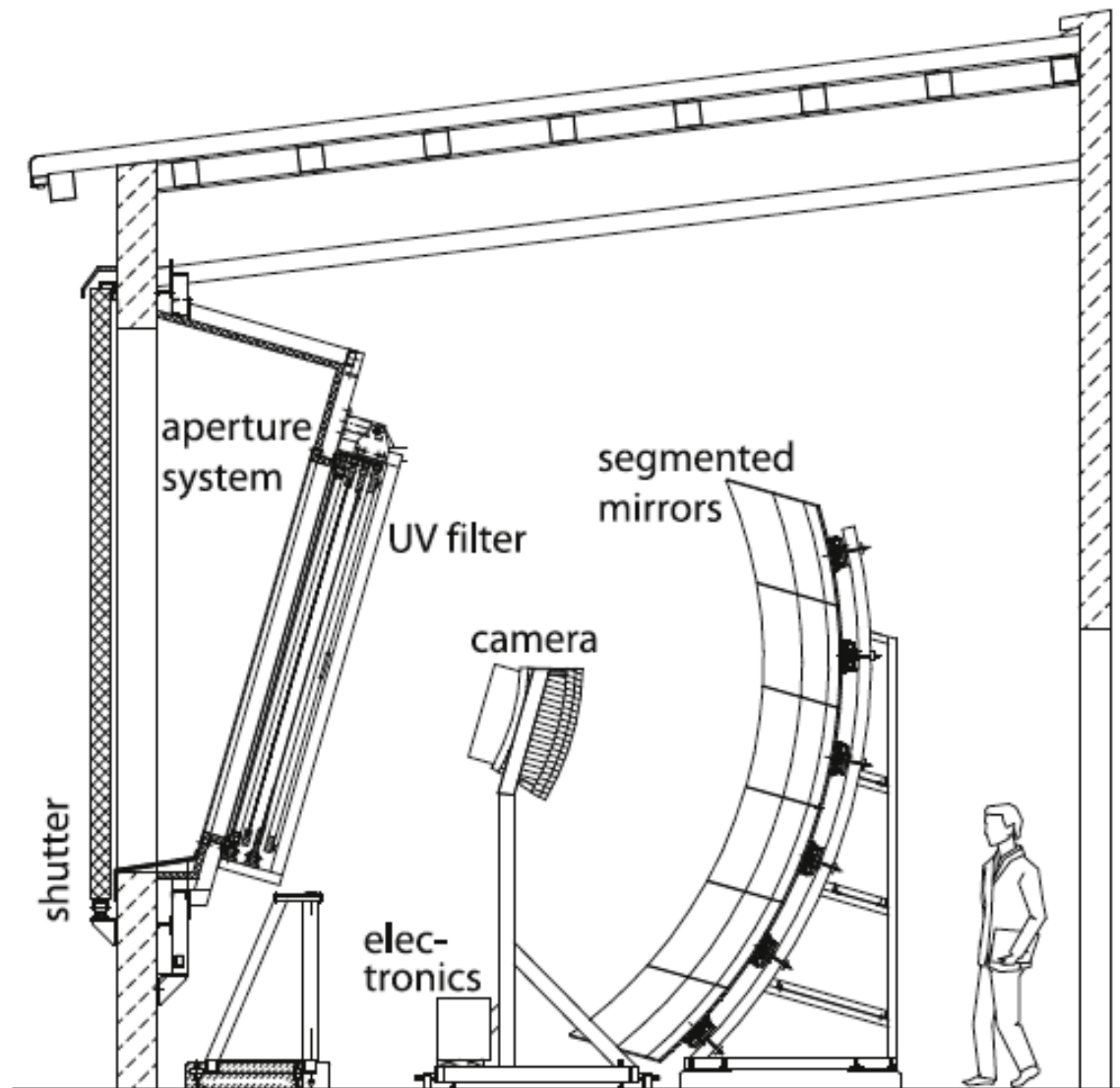


A fluorescence detector



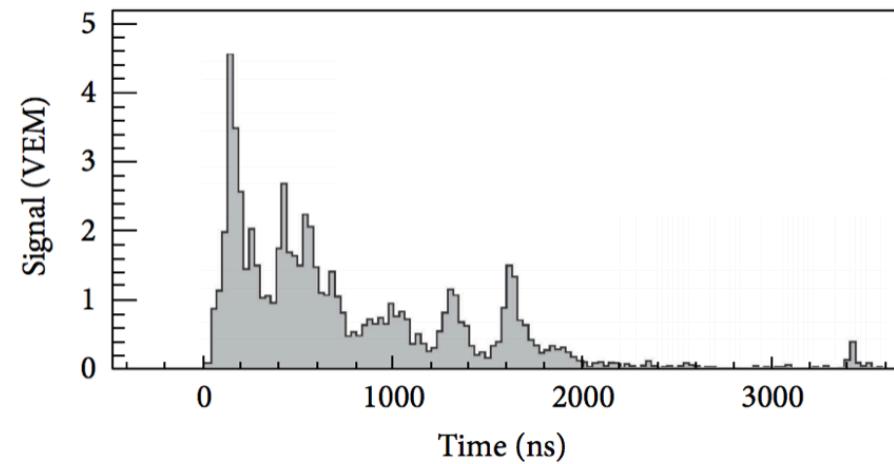
A fluorescence telescope

NIM A 620 (2010) 227–251

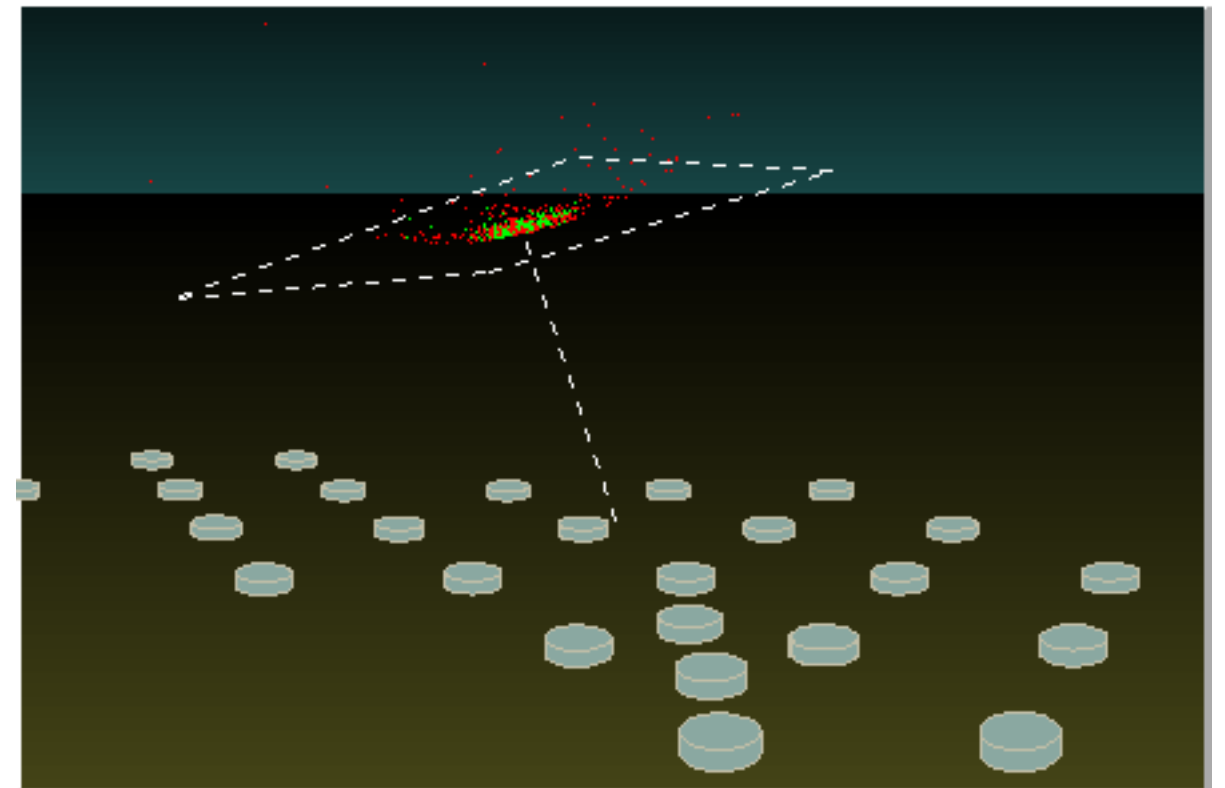
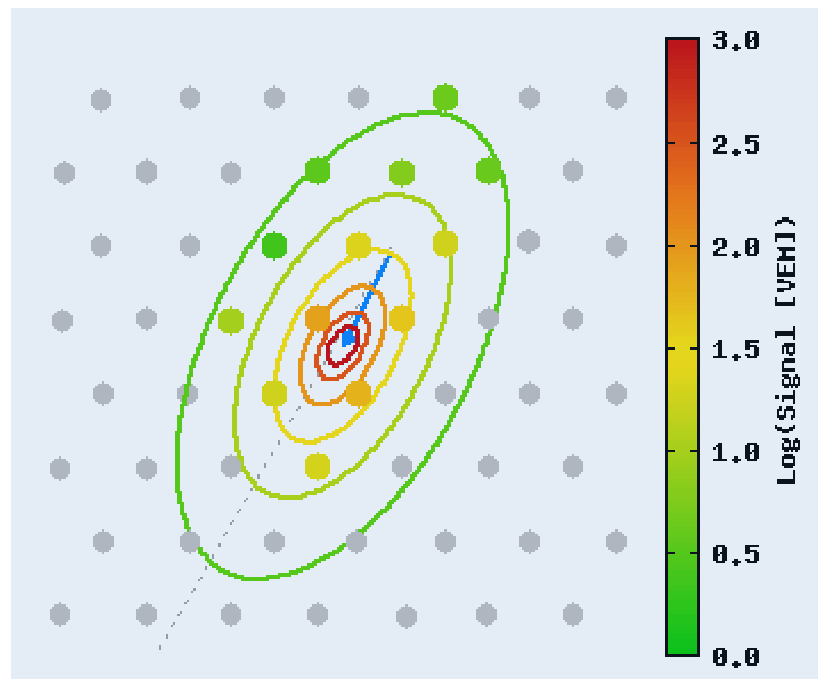


SD event measurement

SD i -th station measures the trigger time t_i and deposited signal S_i



Footprint of the shower on the ground:



Geometric reconstruction (plane shower front)

Determination of:

T_0 : time of impact at ground

u, v : direction cosines

Trigger time of i -th station at position (x_i, y_i) :

$$t_i = T_0 - \frac{u(x_i - x_c) + v(y_i - y_c)}{c}$$

Build a chi2:

$$\chi^2 = \sum_{i=1}^N \left(\frac{t_i^m - t_i}{\sigma_i} \right)^2$$

Uncertainty on trigger time

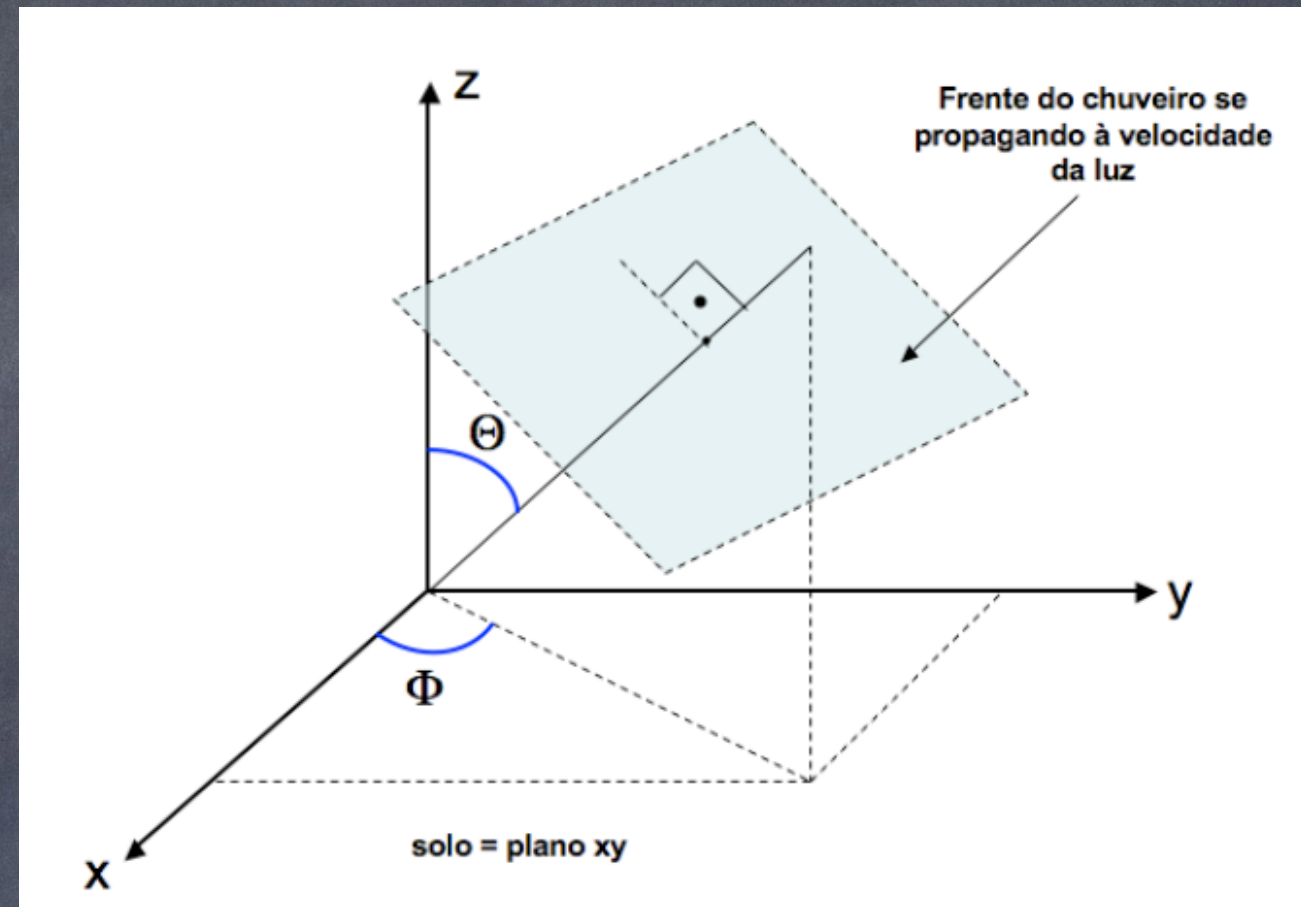
$$\frac{\partial \chi^2}{\partial T_0} = \frac{\partial \chi^2}{\partial u} = \frac{\partial \chi^2}{\partial v} = 0$$

$$(u', v') = (-u/c, -v/c)$$



(Linear system)

$$\underbrace{\begin{pmatrix} \sum_i w_i & \sum_i w_i \bar{x}_i & \sum_i w_i \bar{y}_i \\ \sum_i w_i \bar{x}_i & \sum_i w_i \bar{x}_i^2 & \sum_i w_i \bar{x}_i \bar{y}_i \\ \sum_i w_i \bar{y}_i & \sum_i w_i \bar{x}_i \bar{y}_i & \sum_i w_i \bar{y}_i^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} T_0 \\ u' \\ v' \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} \sum_i w_i t_i^m \\ \sum_i w_i \bar{x}_i t_i^m \\ \sum_i w_i \bar{y}_i t_i^m \end{pmatrix}}_{\mathbf{b}}$$



(x_c, y_c) : ponto de impacto no solo

$$u = \sin \theta \cos \phi$$

$$v = \sin \theta \sin \phi$$

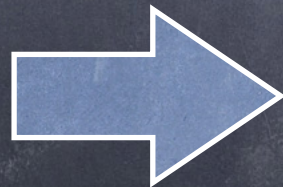
$$w_i = 1/\sigma_i^2$$

Geometric reconstruction (plane shower front)

Stations cannot be all aligned!!!

$$\underbrace{\begin{pmatrix} \sum_i w_i & \sum_i w_i \bar{x}_i & \sum_i w_i \bar{y}_i \\ \sum_i w_i \bar{x}_i & \sum_i w_i \bar{x}_i^2 & \sum_i w_i \bar{x}_i \bar{y}_i \\ \sum_i w_i \bar{y}_i & \sum_i w_i \bar{x}_i \bar{y}_i & \sum_i w_i \bar{y}_i^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} T_0 \\ u' \\ v' \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} \sum_i w_i t_i^m \\ \sum_i w_i \bar{x}_i t_i^m \\ \sum_i w_i \bar{y}_i t_i^m \end{pmatrix}}_{\mathbf{b}}.$$

Aligned stations



$\det(\mathbf{A})=0$ (singular matrix)

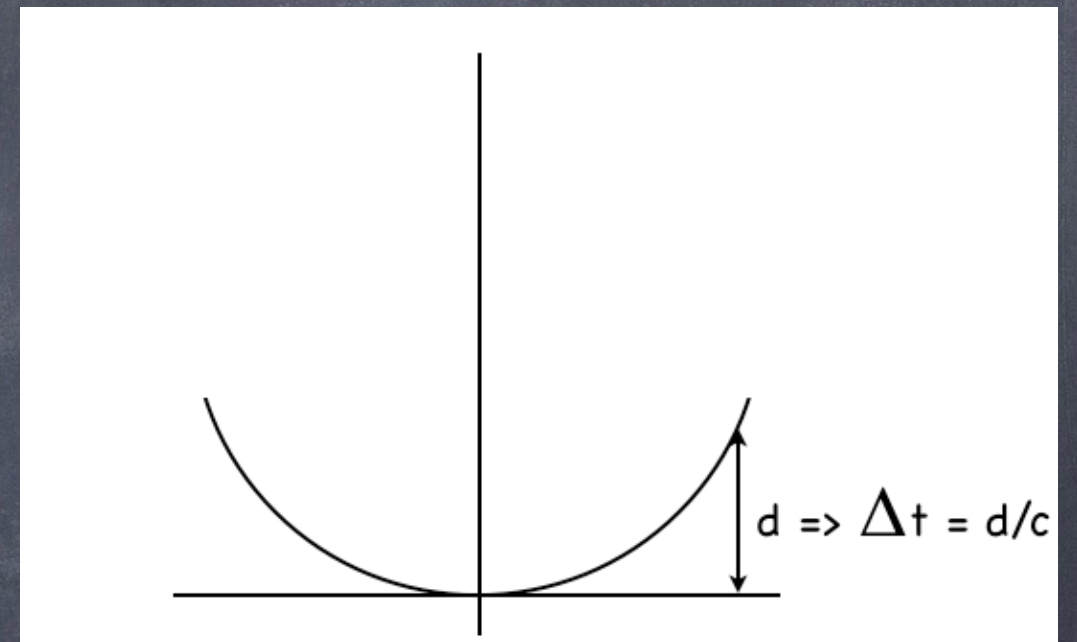
Geometric reconstruction (curved shower front)

Curvature correction for a spherical shower front:

$$\gamma = \frac{1}{2R}$$

$$t_i = T_0 - \frac{u(x_i - x_c) + v(y_i - y_c)}{c} + \gamma (x_i^2 + y_i^2 - (ux_i + vy_i)^2) / c$$

To fit the curvature radius R, one need at least 4 stations



Altitude correction for each station:

$$\delta t_i^{alt} = (z_i - Z_{ref}) \cos \theta / c = (z_i - Z_{ref}) (1 - u^2 - v^2)^{1/2} / c$$

At the end of the process, one has the following parameters: T_0 , θ , ϕ and R

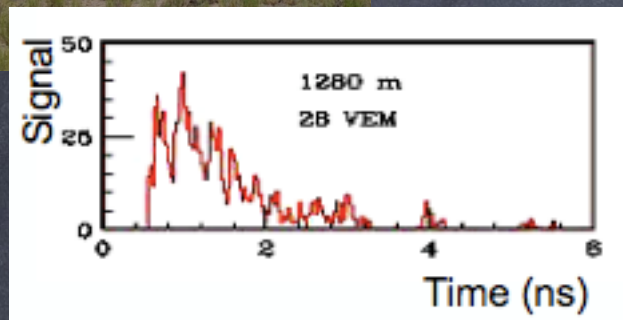
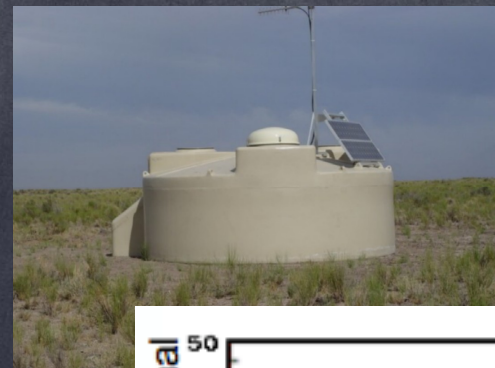
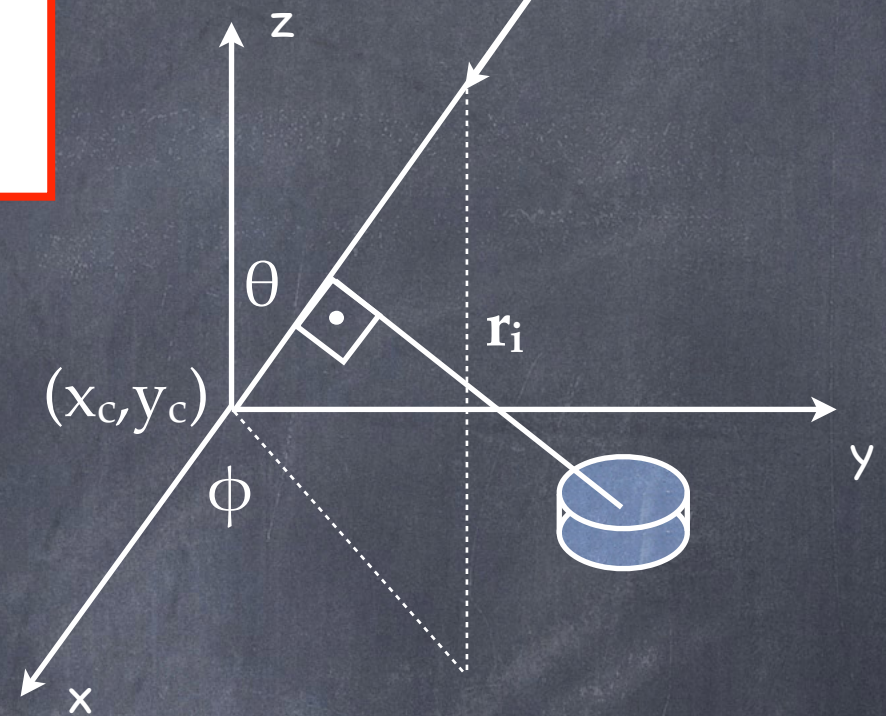
Lateral Distribution Function (LDF) reconstruction

Model for the shower lateral profile

$$\rho(r) = S(1000) \left[\frac{r}{r_0} \left(\frac{r + r_s}{r_0 + r_s} \right) \right]^\beta$$

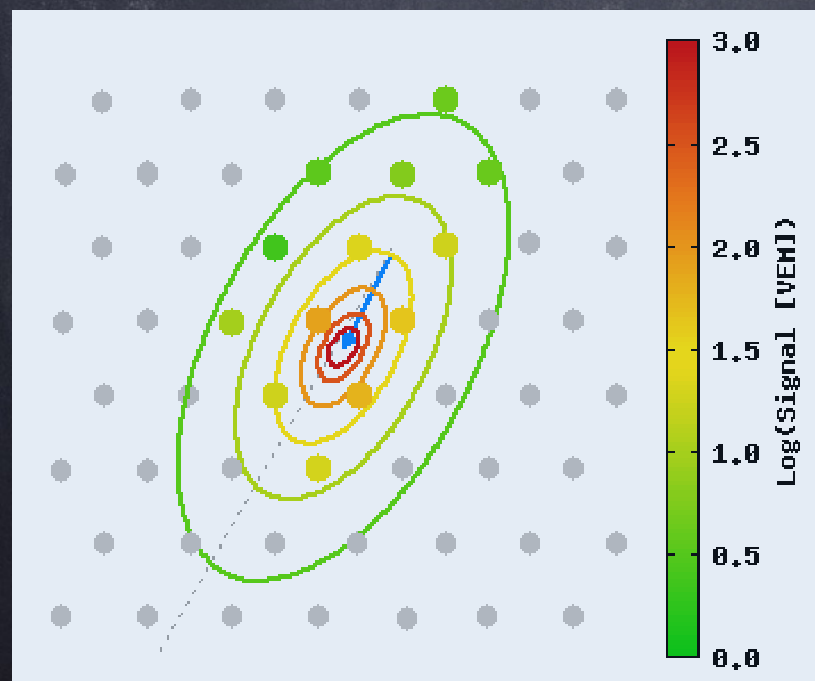
$$r_0 = 1000 \text{ m}$$

$$r_s = 700 \text{ m}$$

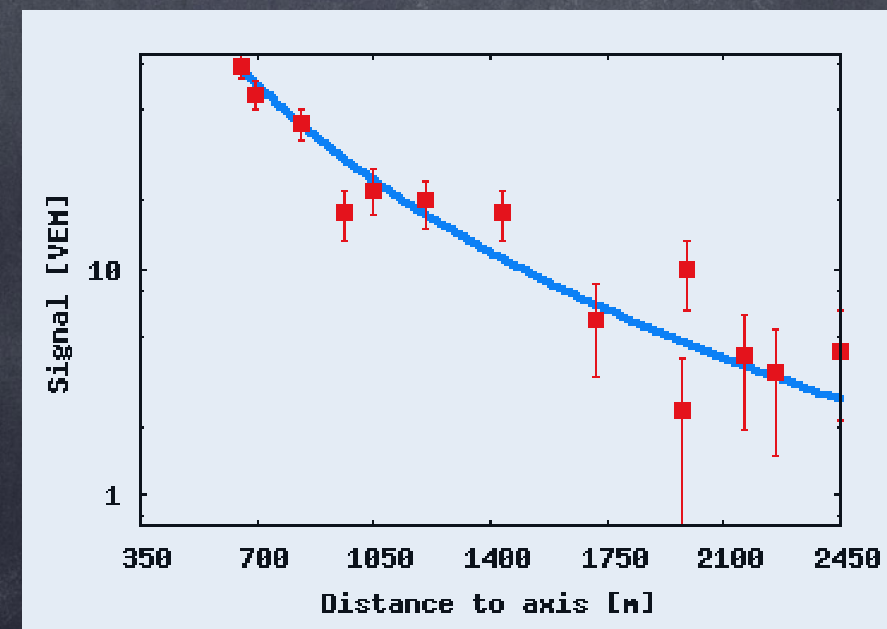


LDF (Lateral Distribution Function)

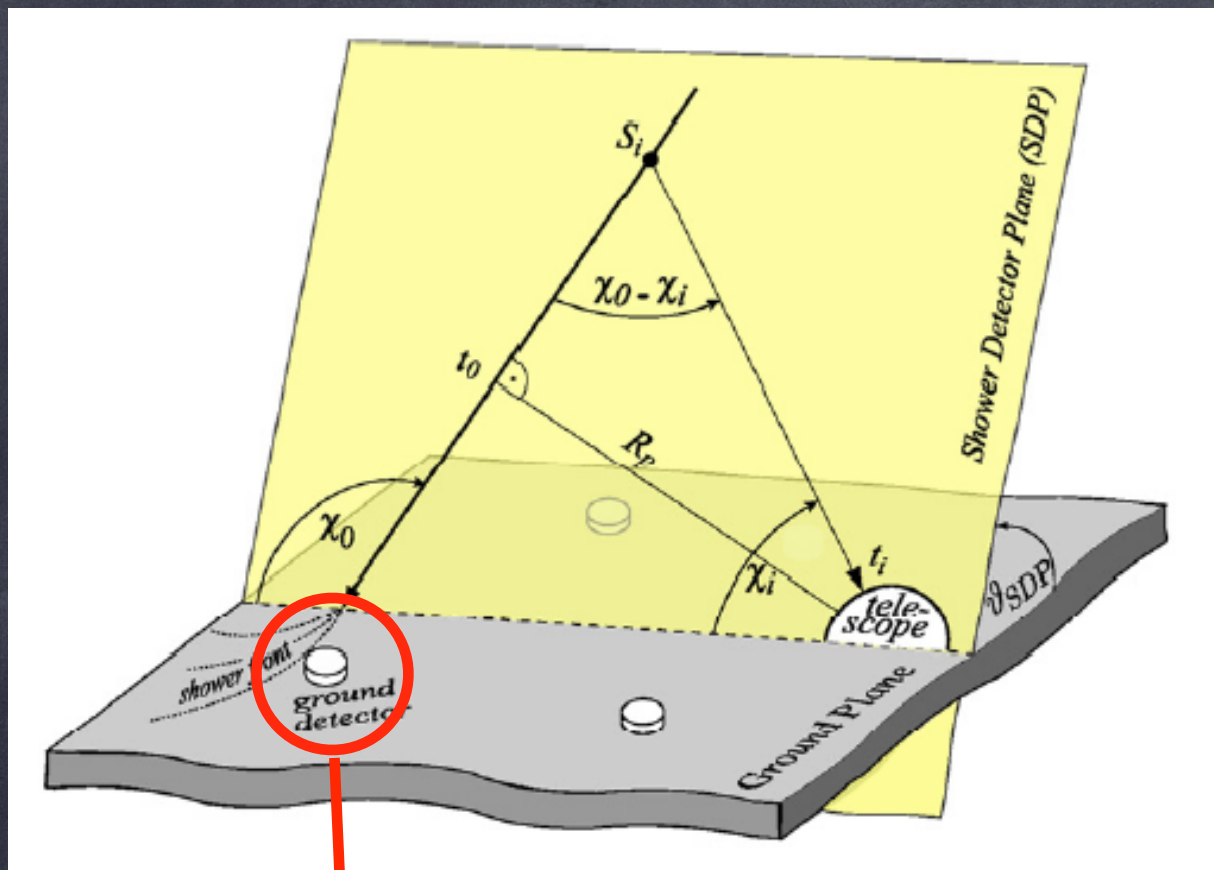
Shower footprint



Model fit to the data



Hybrid geometric reconstruction



Trigger time of i-th pixel of the FD camera:

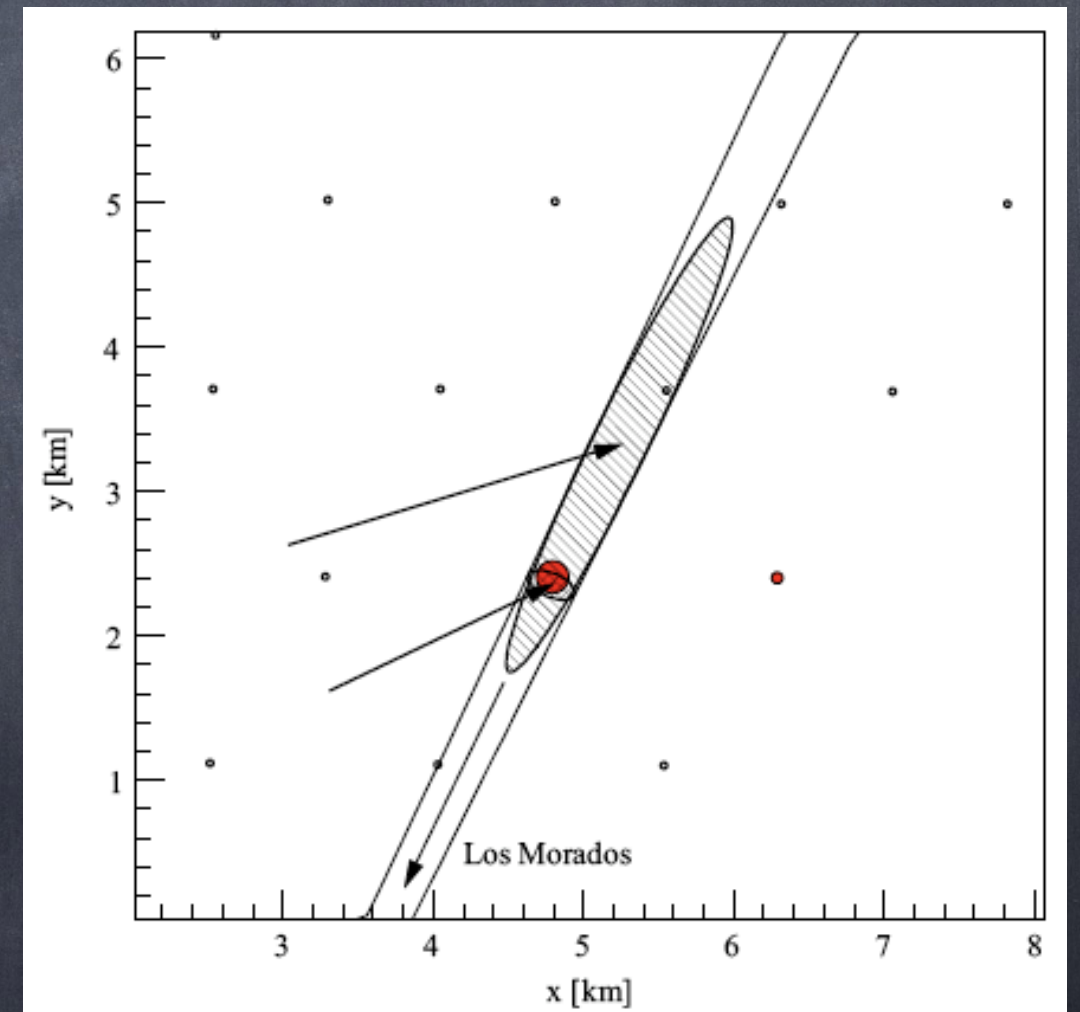
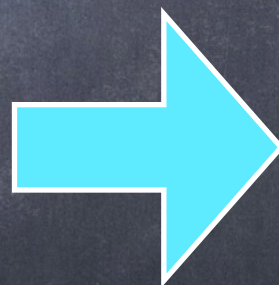
$$t_i = t_0 + \frac{R_p}{c} \tan \left(\frac{\chi_0 - \chi_i}{2} \right).$$

+ single tank at ground:

$$t_{\text{tank}} = \frac{ux_{\text{tank}} + vy_{\text{tank}}}{c}$$

$$u = \sin \theta \cos \phi$$

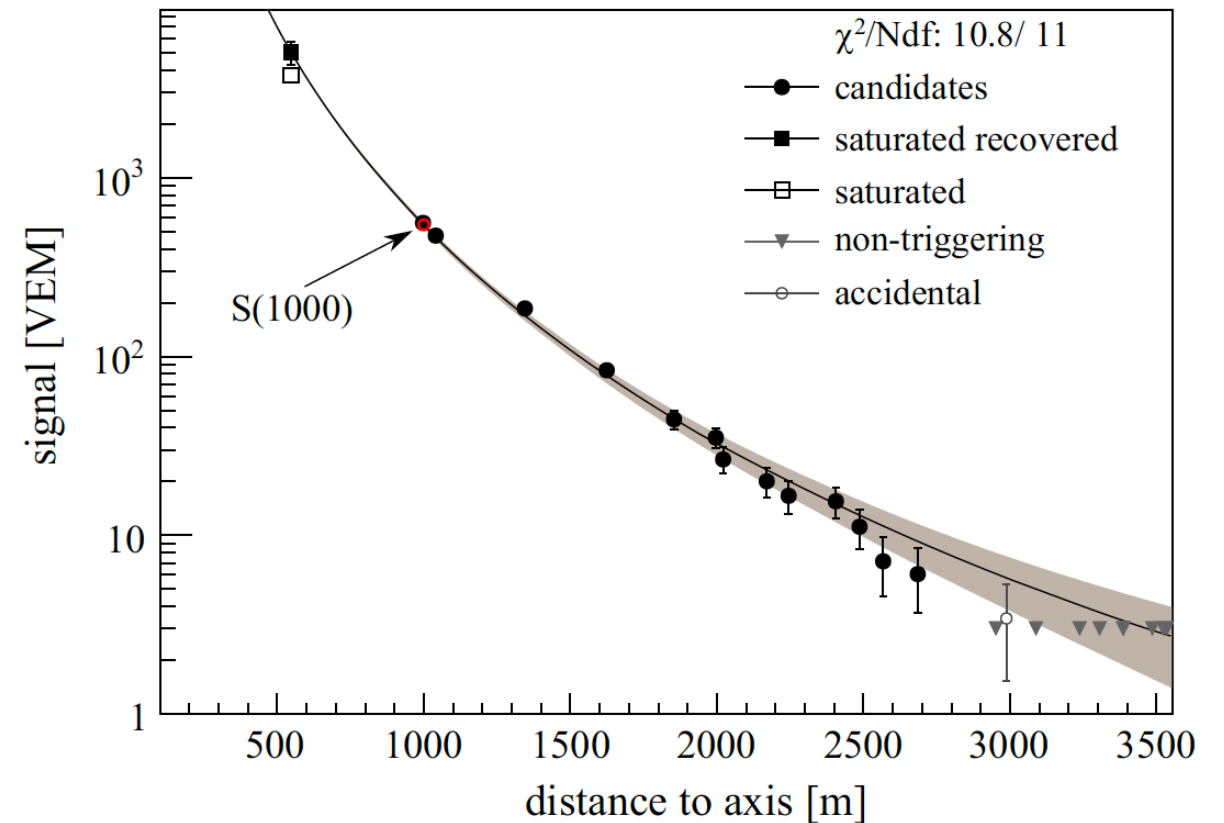
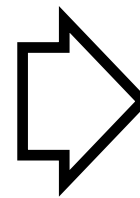
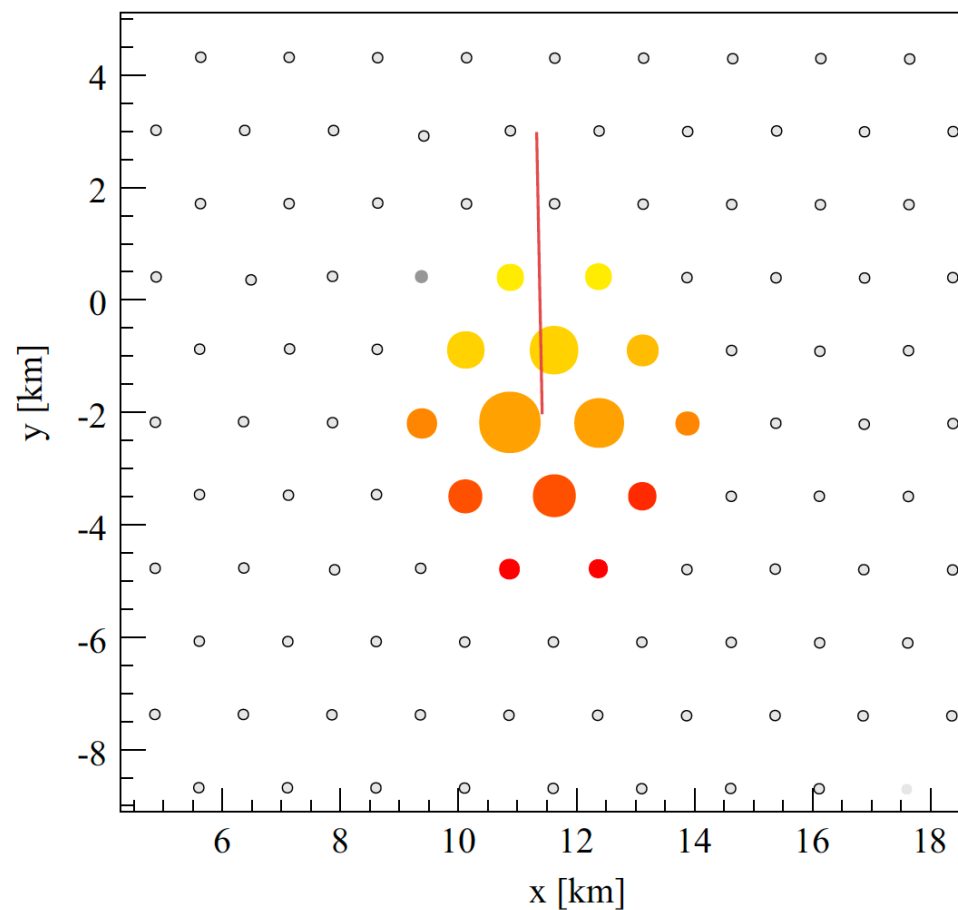
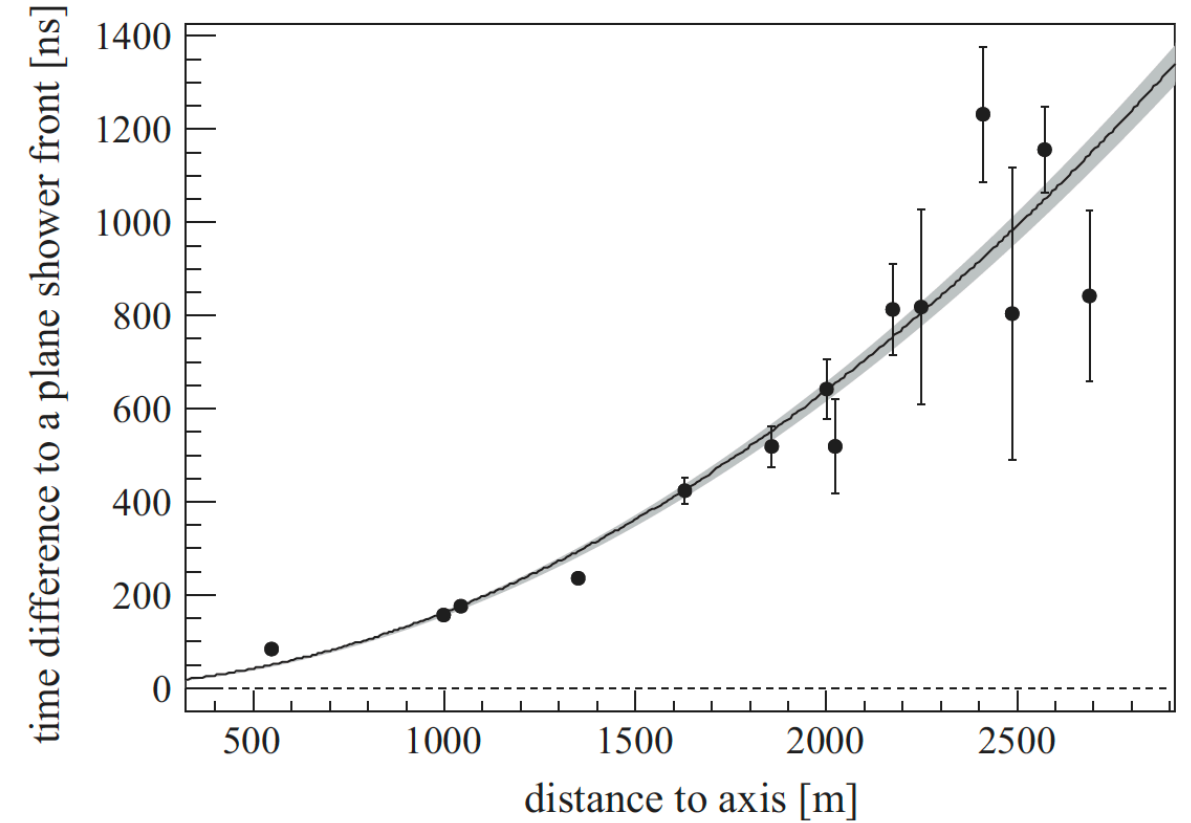
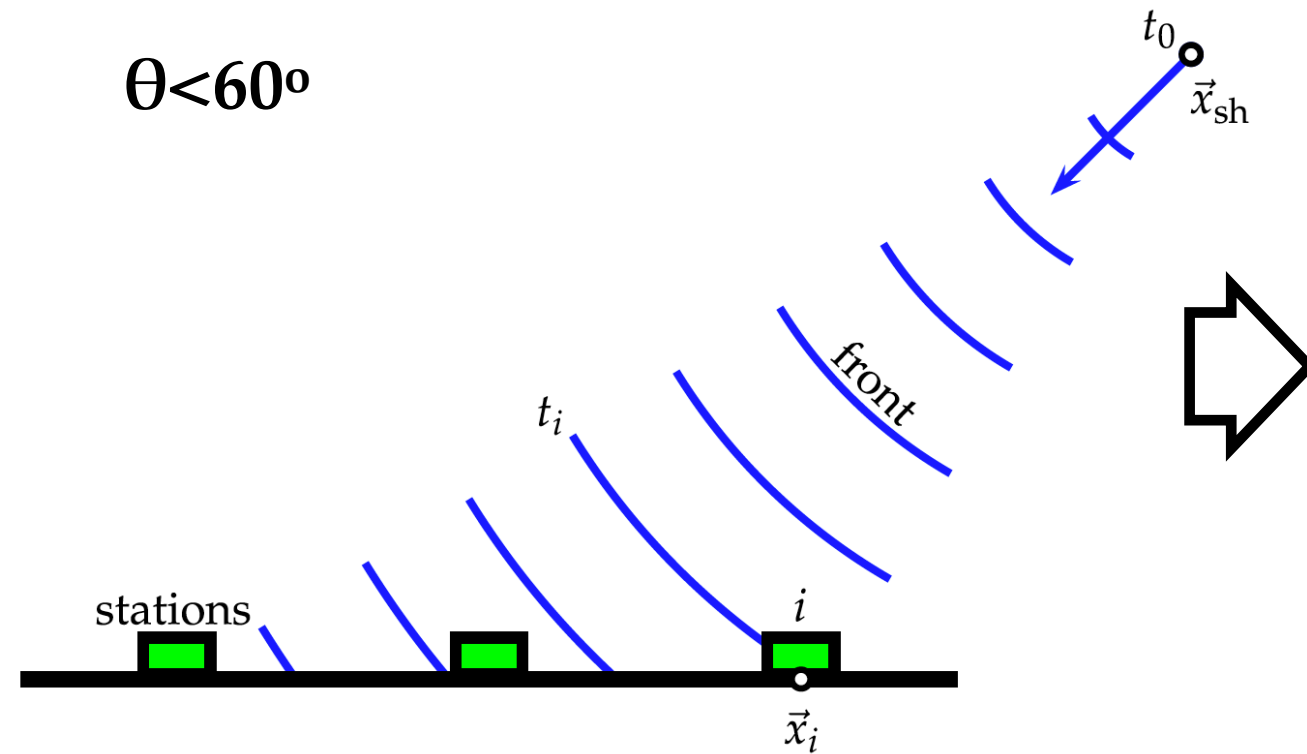
$$v = \sin \theta \sin \phi$$



Angular resolution better than 1 deg

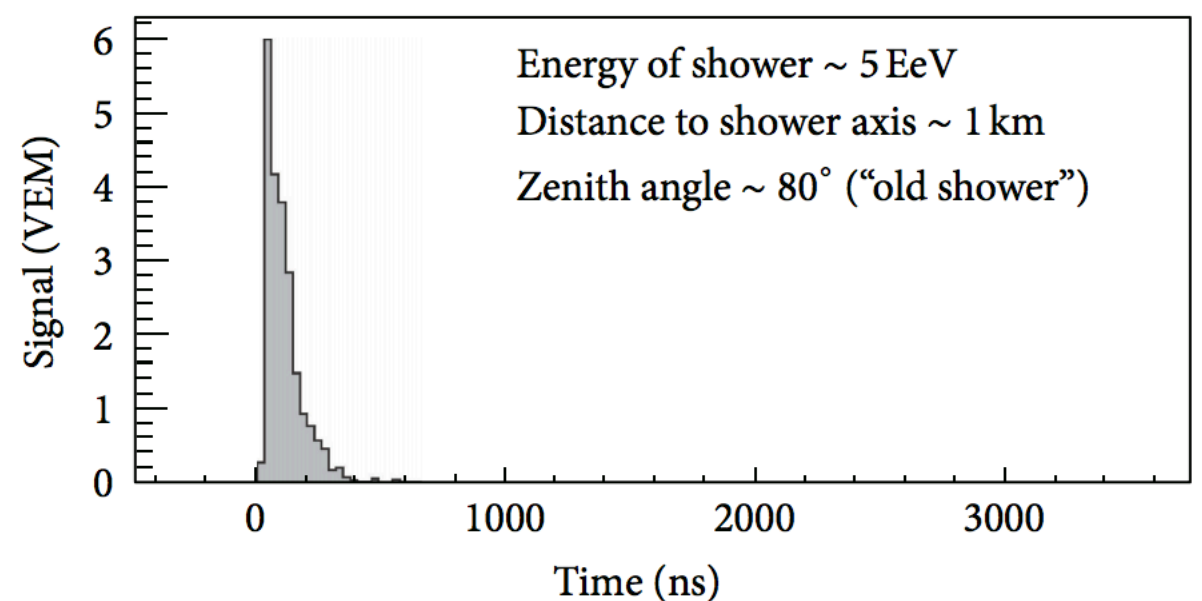
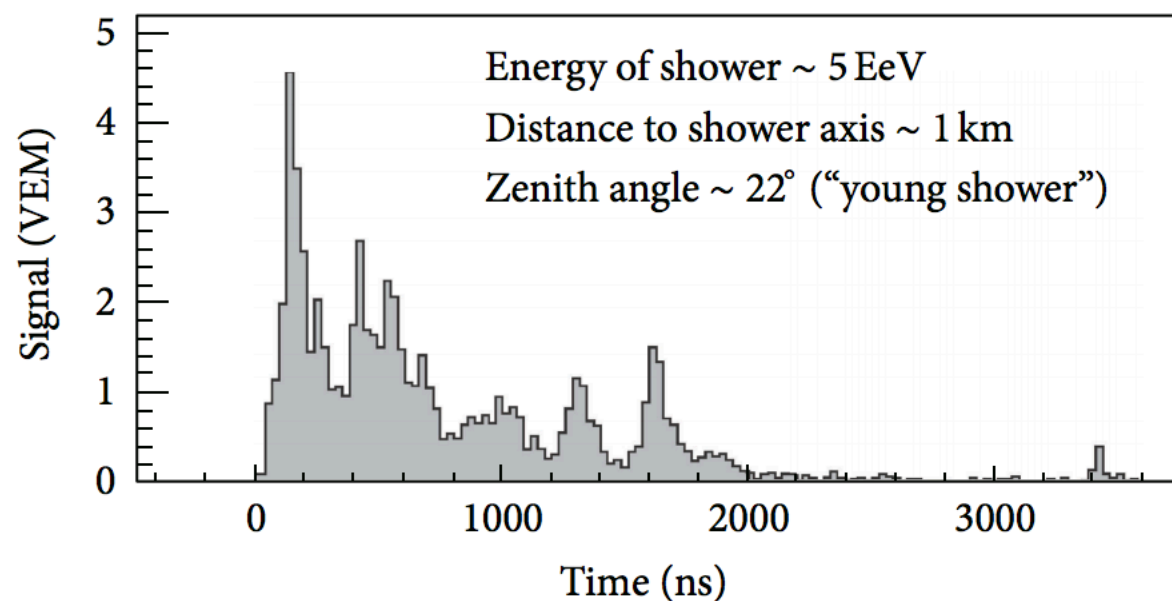
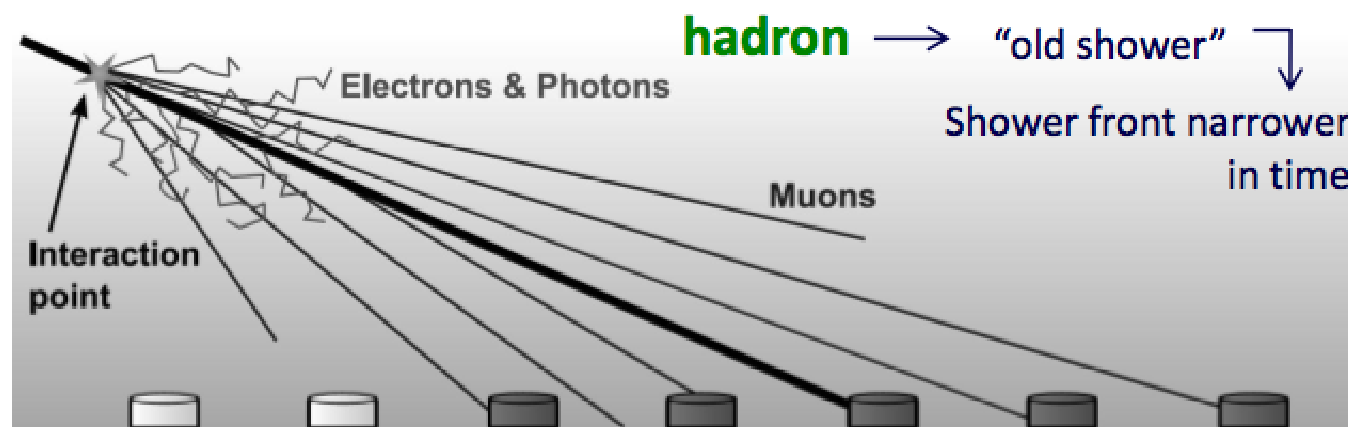
Excellent resolution on the shower
core impact point

Surface detector vertical event reconstruction



Inclined showers

- Showers developing at large zenith angle are strongly affected by atmospheric attenuation
- Signals at ground are muon rich
- Lateral distribution density of particles can no longer be described by a continuous function of the distance to shower axis



Inclined showers

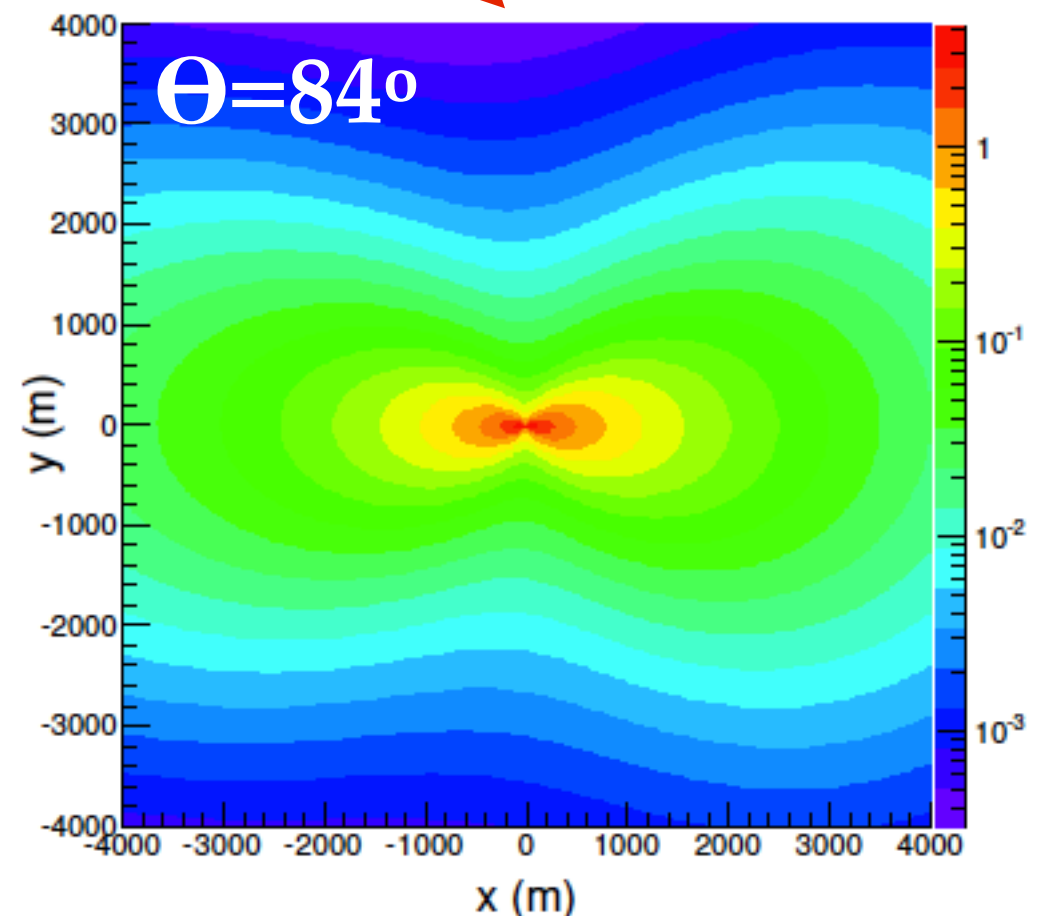
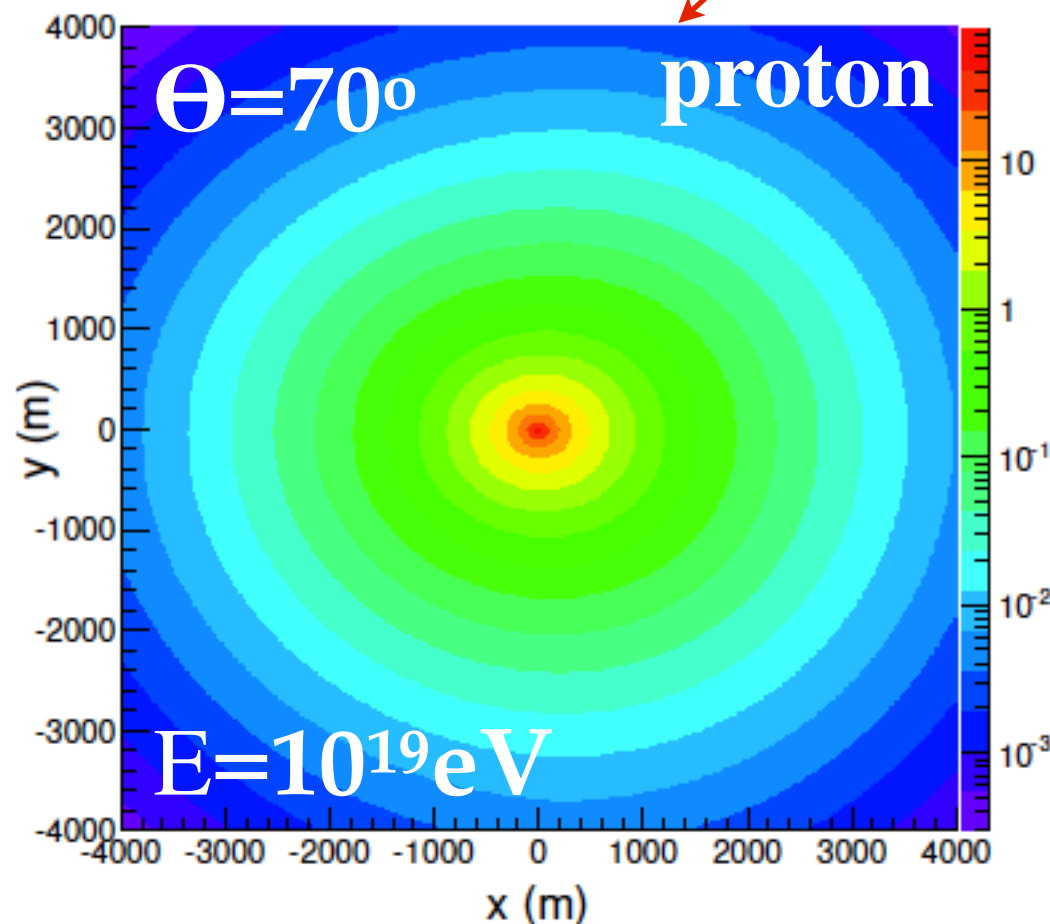
- EAS muon content depends on: **energy, primary particle, zenith and azimuth**

primary, had. model, energy, geometry

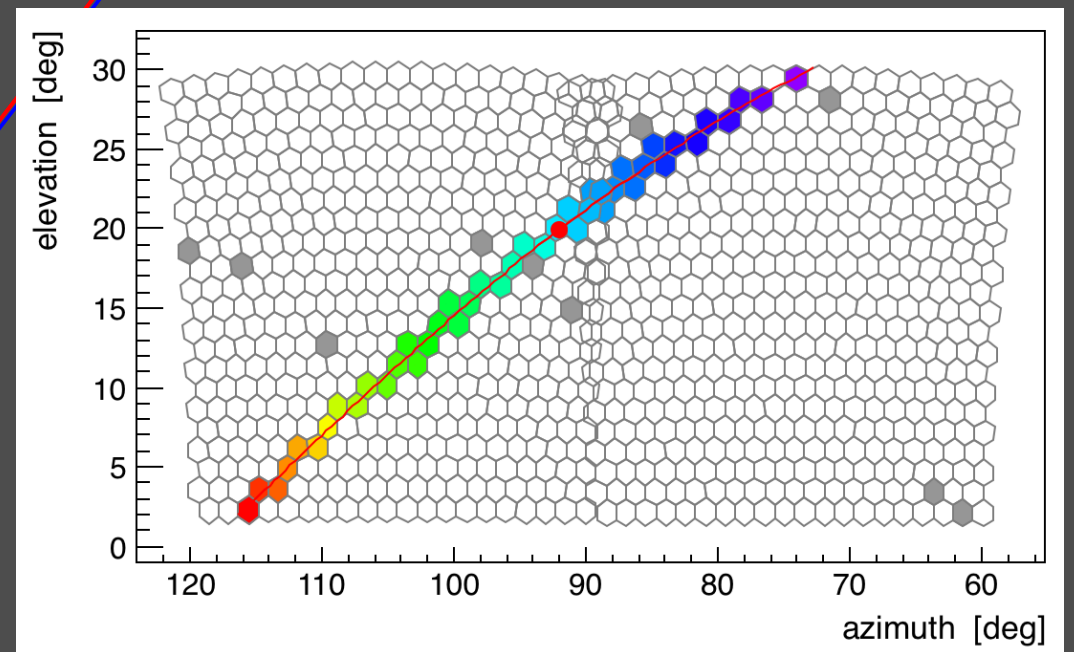
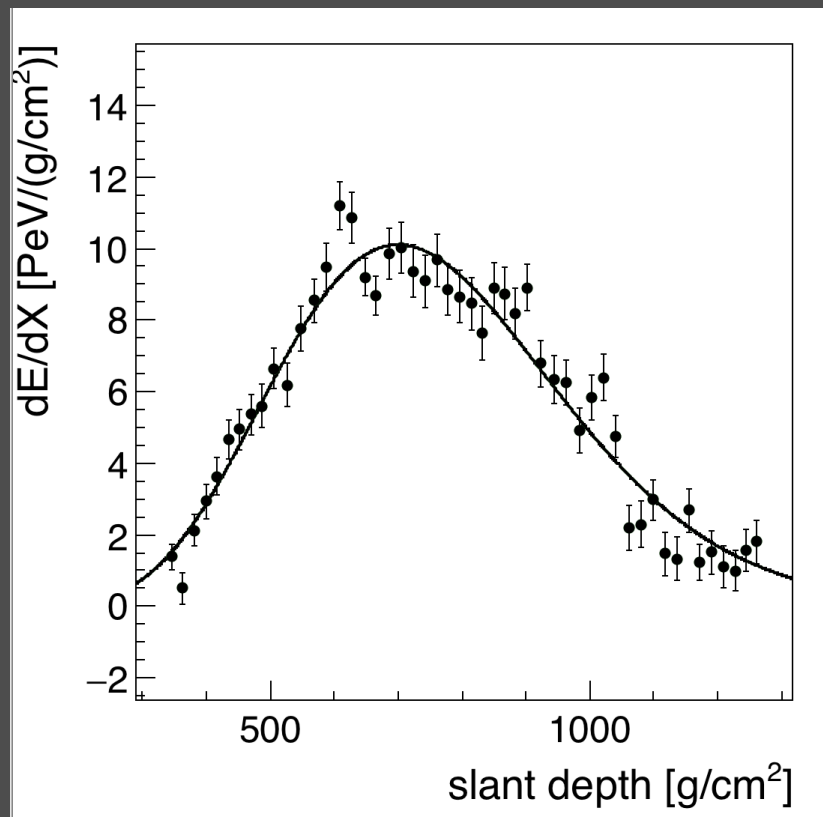
reference muon density map

$$\rho_{\mu}(\vec{r}) = N_{19} \rho_{\mu,19}(\vec{r}; \theta, \phi)$$

measured muon density



Hybrid event

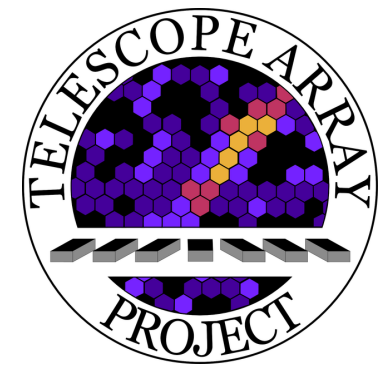


$$X = \int \rho dx \quad (\text{g}/\text{cm}^2)$$

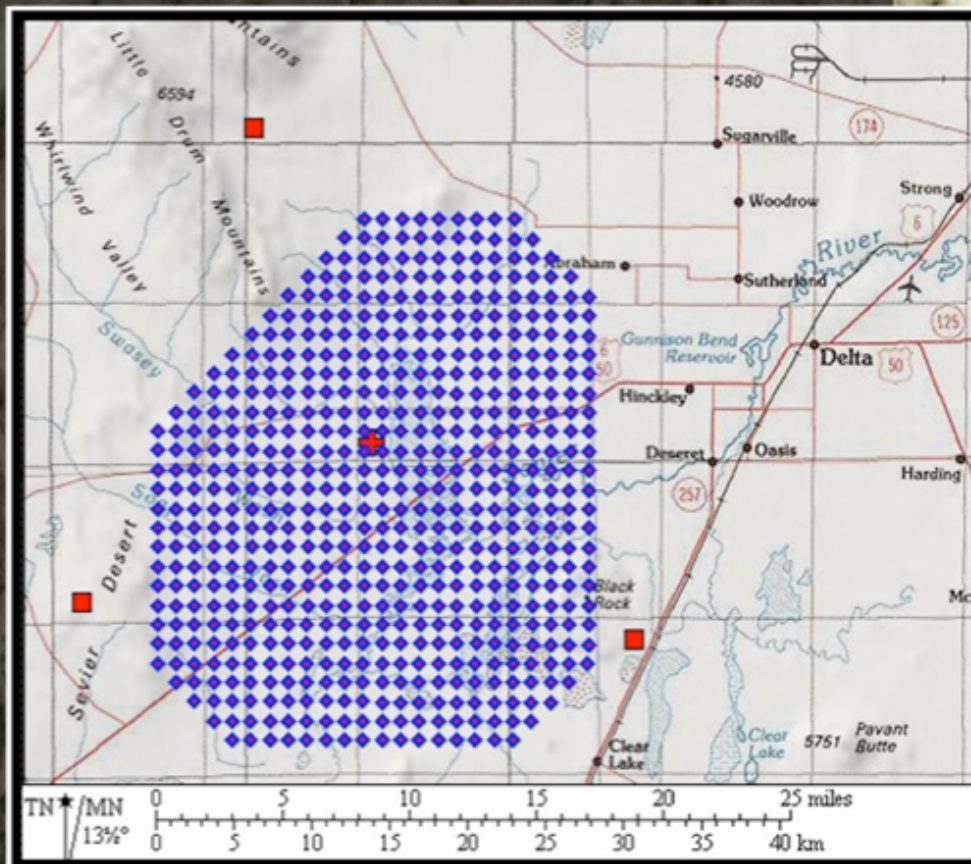
$$N_e(X) = N_{max} \left(\frac{X - X_0}{X_{max} - X_0} \right)^{\frac{X_{max} - X_0}{\lambda}} e^{\frac{X_{max} - X_0}{\lambda}}$$



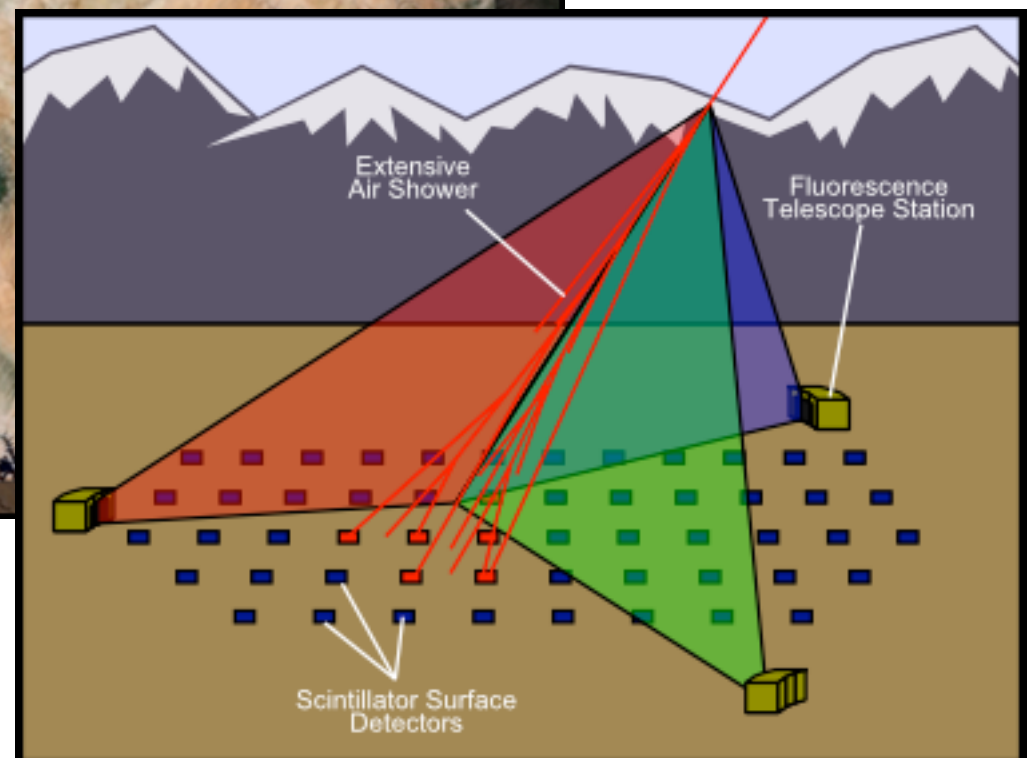
Telescope Array (TA)



~700 km² de área
507 cintiladores plásticos
rede quadrada de 1.2 km
3 telescópios de fluorescência
operando desde 2008

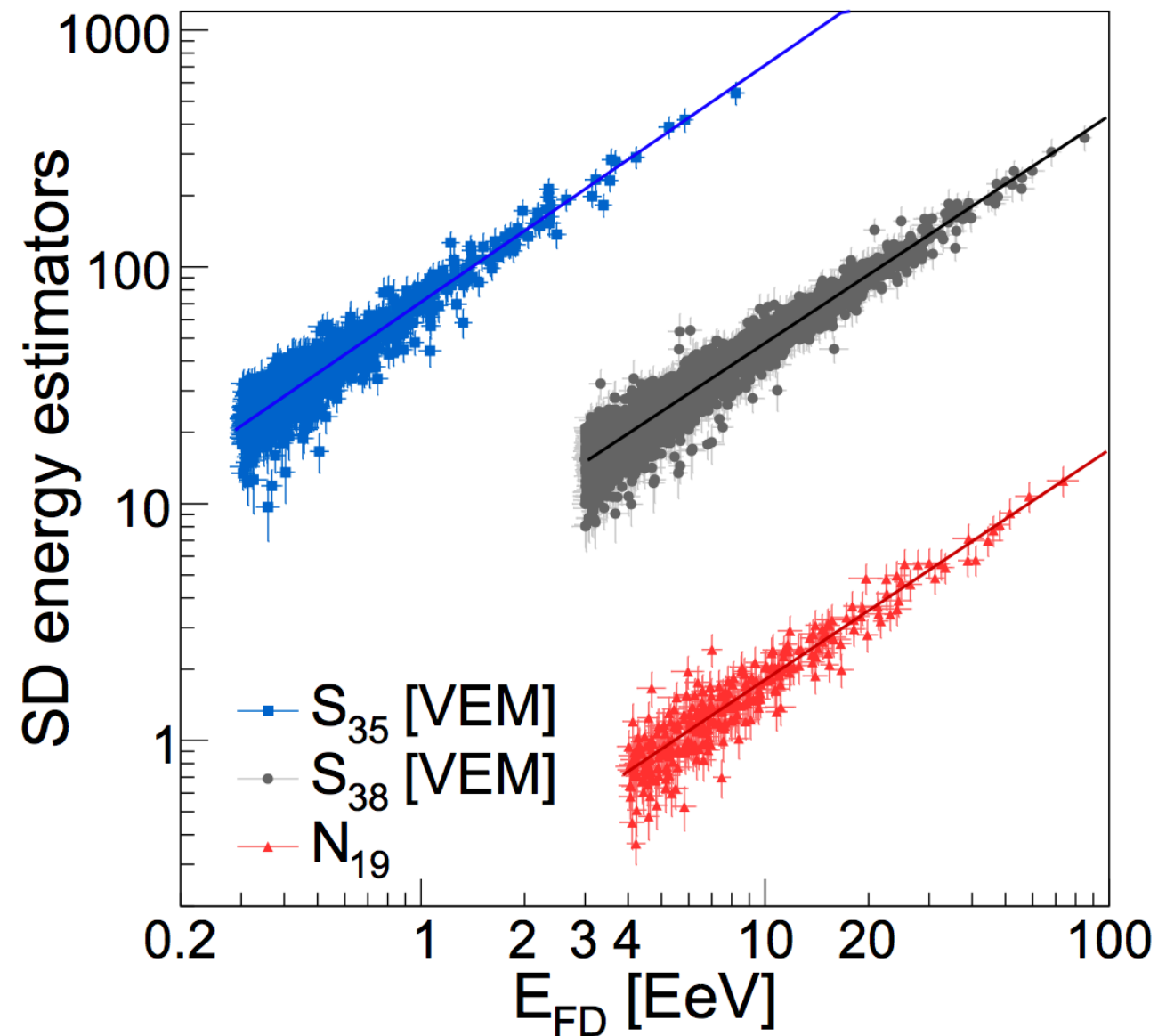


latitude = 39.3°

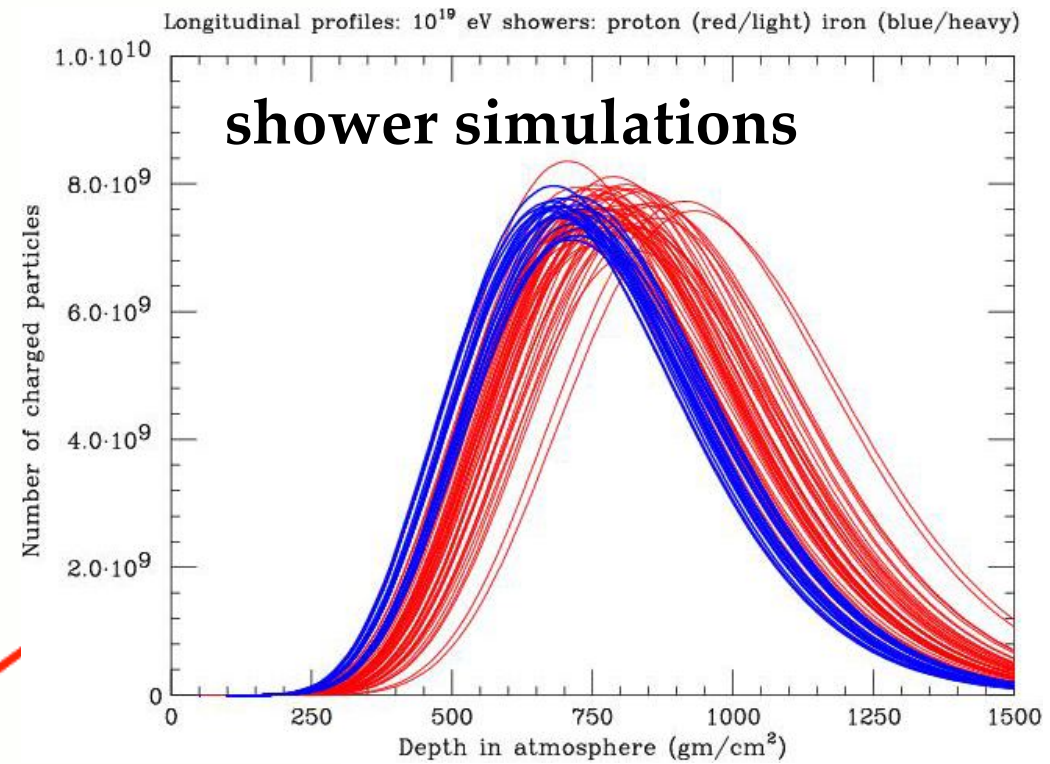


General note on Auger SD data samples - I

- Data-driven energy calibration using hybrid events
- Different SD estimators are correlated to the quasi-calorimetric energy measured by the FD
 - **Vertical: $0^\circ < \theta < 60^\circ$ ($S_{38} \times E_{FD}$)**
 - Vertical: $0^\circ < \theta < 60^\circ$ ($S_{35} \times E_{FD}$)
 - **Inclined: $60^\circ < \theta < 80^\circ$ ($N_{19} \times E_{FD}$)**



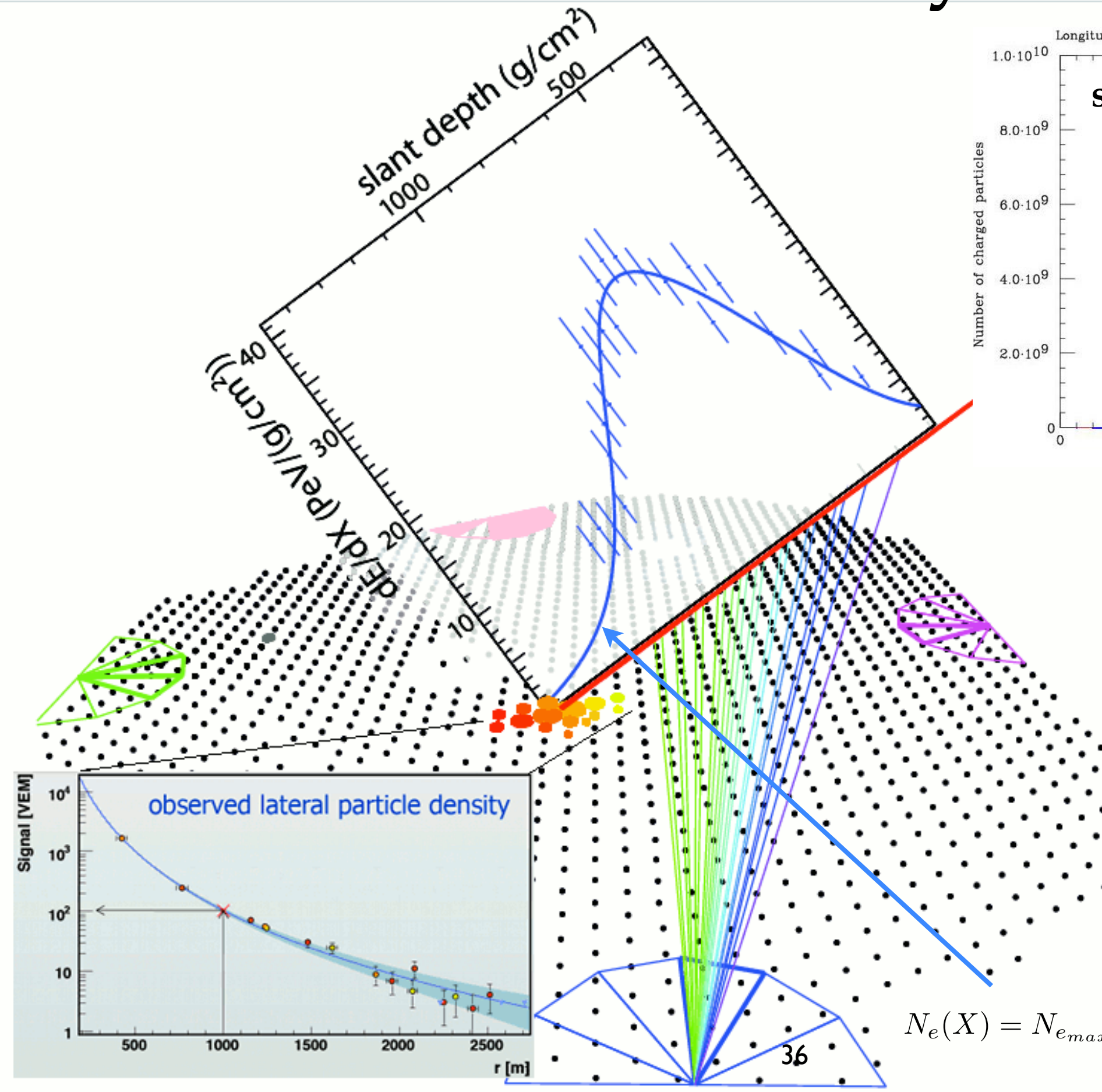
Back to detection in hybrid mode...



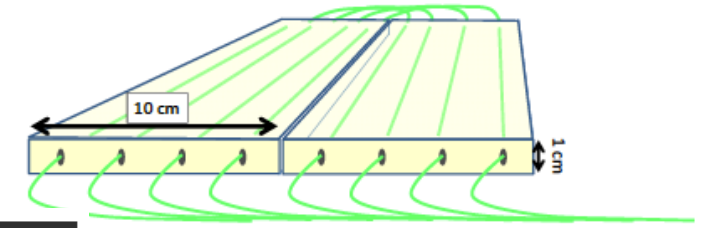
Atmospheric depth at
which shower is maximum
is sensitive to mass number

Gaisser-Hillas profile:

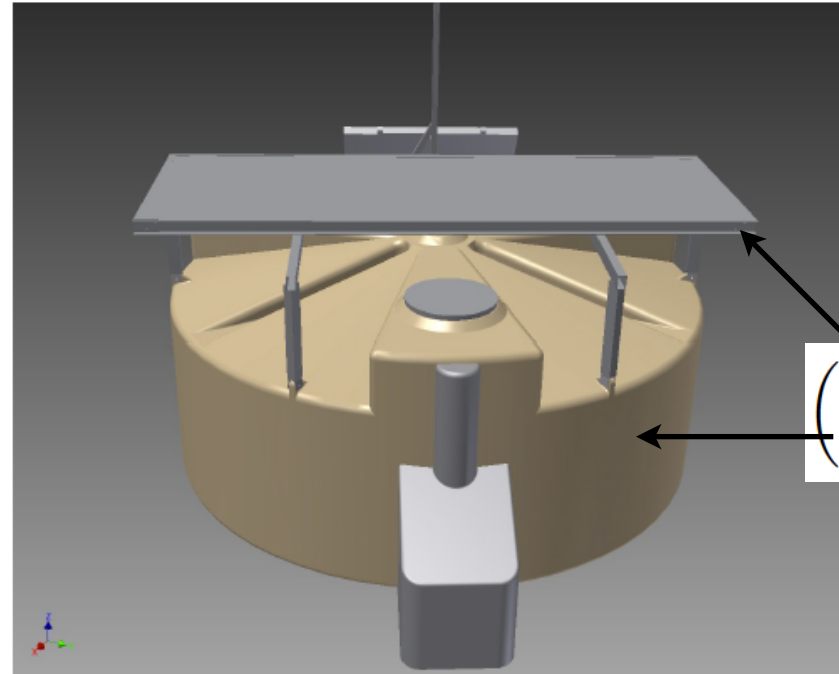
$$N_e(X) = N_{e_{max}} \left(\frac{X - X_0}{X_{max} - X_0} \right)^{\frac{X_{max} - X_0}{\lambda}} e^{\frac{X_{max} - X_0}{\lambda}}$$



AugerPrime



- SSD (Surface Scintillator Detector)
for improved measurement of electromagnetic to muon abundance
- UMD (Underground Muon Detector)
for direct counting of muons



$$\begin{pmatrix} S_{\text{SSD}} \\ S_{\text{WCD}} \end{pmatrix} = \begin{pmatrix} \lambda \mathcal{A}_{\text{SSD}} & \mathcal{A}_{\text{SSD}} \\ \beta \mathcal{A}_{\text{WCD}} & \mathcal{A}_{\text{WCD}} \end{pmatrix} \begin{pmatrix} \mathcal{F}_{\text{em}} \\ \mathcal{F}_{\mu} \end{pmatrix}$$

- SDEU (SD Electronics Upgrade)
faster sampling of the PMT signals (40 MHz to 120 MHz),
better timing accuracy (new GPS) and larger dynamic range
- Enhanced FD duty cycle
15% to 25% increase in the FD operation time
by operating with lower PMT gain

